Lecture notes for EUGLOH workshop in Paris "Physics for Biology" for undergraduate physics students

Tobias Ambjörnsson

1 June – 4 June 2026

Contents

1 Introduction

Living systems are among the most complex forms of organized matter found in nature. Even a single cell contains thousands of interacting molecular species, continuously consuming energy, processing information, transporting cargo, changing shape, and responding to its environment.

At first sight, this can look completely overwhelming. A cell is not a pendulum, not an ideal gas, not a harmonic oscillator, and not a simple electric circuit. It is wet, noisy, crowded, active, chemically complex, and very much alive.

So why should physics be useful?

The answer is that living systems, however complicated, still have to obey physical laws. Molecules diffuse driven by thermal kicks. Fluids exert viscous drag following the equations in fluid mechanics. Statistical physics relate energies to probabilities and constrain what cells can and cannot do.

A large part of biophysics is therefore not about memorizing biological details. It is about learning how to ask physical questions about biological systems:

Which effects are large/small?   What is the simplest useful physics model?

This is the spirit of these notes.

Cells operate on length scales ranging from nanometers to micrometers. At these scales, thermal fluctuations become important, diffusion often dominates transport, and viscous forces dominate over inertia. As a consequence, microscopic organisms effectively inhabit a physical world very different from our everyday macroscopic experience.

For example, a bacterium swimming in water experiences a Reynolds number many orders of magnitude smaller than that of a fish or a human swimmer. When propulsion stops, motion ceases almost immediately. There is no coasting. Similarly, molecular motion inside cells is strongly influenced by random thermal fluctuations, making stochastic descriptions unavoidable.

Another important feature of living systems is that they often operate far from thermodynamic equilibrium. Cells continuously consume energy, primarily through ATP hydrolysis, in order to maintain organization, generate forces, transport cargo, and perform mechanical work. Biological systems are therefore intrinsically active systems.

Modern biophysics attempts to understand how relatively simple physical principles can give rise to the remarkable functionality observed in living matter. In many cases, surprisingly simple models capture essential features of complex biological processes. We will encounter several such simplified models.

Another important skill if you want to do biophysics is to use of simple physical estimates. Before constructing a detailed model, it is often useful to ask basic quantitative questions:

- Is diffusion sufficiently fast for this process?
- Which force dominates: inertia, viscosity, elasticity, or thermal fluctuations?
- Is an energy difference large or small compared with $k_B T$?
- What is the relevant time scale?

- Which effects can safely be neglected?

Such estimates are not just warm-up exercises. They are often the most important part of the physics. They tell us what kind of model is worth writing down in the first place.

The notes combine analytical calculations with simple stochastic simulations and numerical methods. The goal is not to turn the workshop into a programming course, but we will repeatedly use the computer for a fast way to explore the models we introduce. If we write down a random walk, we should also be able to simulate one. If we derive a rate equation, we should also be able to integrate it numerically. If we estimate a diffusion constant from data, we should understand what statistical assumptions went into that estimate. In this sense, the notes are written from the point of view that physics and computation belong together:

$$\text{model} \quad \longrightarrow \quad \text{equation} \quad \longrightarrow \quad \text{simulation} \quad \longrightarrow \quad \text{physical interpretation.}$$

What you should learn from this workshop

By the end of the workshop, you should not be an expert in all of biophysics. That is not the goal. The goal is instead that you should have a useful physical toolkit for thinking about living systems.

More concretely, after the lectures and projects you should be able to:

- explain why diffusion is efficient over microscopic distances but inefficient over macroscopic distances;
- derive the diffusion equation from a simple random-walk model;
- simulate Brownian motion trajectories using Gaussian random numbers;
- understand the likelihood function for Brownian displacements and use it to infer a diffusion constant from trajectory data;
- simulate and analyze first-passage or capture processes, such as the time it takes a diffusing particle to reach an absorbing target;
- understand why low Reynolds number motion is dominated by viscosity rather than inertia;
- derive and interpret simple hydrodynamic results such as Poiseuille flow;
- explain the Stokes friction law and the Einstein relation;
- describe molecular motors as stochastic, nonequilibrium machines that convert chemical energy into directed motion;
- model a motor protein as a biased random walker and compute its mean velocity and effective diffusion coefficient;
- write down and interpret rate equations for irreversible, reversible, bimolecular, and enzymatic reactions;

- derive the Michaelis–Menten equation from a simple enzyme reaction scheme and solve the equations numerically using Euler's method.

Equally important are the more general skills of importance in biophysics. You will, maybe for the first time in your education, spread your wings and try to tackle tasks like:

- making order-of-magnitude estimates,
- translating a biological cartoon into a mathematical model,
- making dimensional arguments,
- using simulation to test intuition,
- deciding which details are essential and which can be ignored.

These are useful skills far beyond the particular examples in the notes.

About these notes

These lecture notes provide an introduction to some of the physical ideas that repeatedly appear in modern cell and molecular biophysics. They are intended primarily for undergraduate physics students with a background in standard undergraduate physics, but with limited prior exposure to biophysics or stochastic processes.

The overall goal is not to provide a complete treatment of biophysics. Instead, the goal is to illustrate how physical reasoning can provide insight into living systems.

The notes are based on excerpts from two courses I am/was teaching at Lund University (Sweden): *Theoretical Biophysics (7.5 ECTS)* and *Computational Physics (7.5 ECTS)*. The cropped handwritten figure are from the Theoretical biophysics course (sorry about the extra lines/boxes etc which appear in some of these figures). The notes were also inspired in part by the book

Phillips, Kondev, Theriot & Garcia,
Physical Biology of the Cell, 2nd ed.

Students interested in further reading are strongly encouraged to consult this textbook, which covers most material a student who wish to pursue masters studies in biophysics should know. The book contains many biological examples, physical arguments, and exercises.

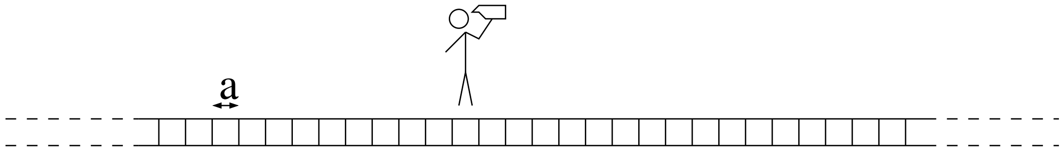
Some of the sections/subsections of these notes are marked with a start (*), these can be omitted at a first reading and are not part of the core material for the workshop.

2 Random numbers and processes

Randomness is a key component in all of biophysics. We here slowly approach random/stochastic processes. We also introduce probability density functions (PDFs), and encounter a particularly important PDF – the *Boltzmann distribution* which underlies all of the field of statistical mechanics.

2.1 Our first random process — where is that drunk student?

Consider a random walker (say, a drunk student walking the streets of Paris) on a one-dimensional lattice (a long cobble-stone covered pavement).



The walker starts at lattice site 0 and takes steps to the left and right with equal probability,

$$p_{\text{left}} = p_{\text{right}} = \frac{1}{2}.$$

The time between jumps is denoted by Δt, and the lattice constant by a.

This process is the simplest possible example of a *stochastic process.* Unlike a system described by Newton's equations of motion, the future position of the random walker cannot be predicted exactly from the initial condition. Even if we know that the walker starts at the origin, we do not know where the walker will be after, say, 100 steps.

Does this mean that the problem is hopeless?

No. It simply means that we must ask statistical questions. For example, if many walkers are prepared in the same initial state, we can ask:

- What is the probability that the walker is at position x at time t?
- What is the average distance from the starting point?
- How long does it take, on average, before the walker reaches a certain point, for instance the student dorm?

Such averages over many independently prepared systems are called *ensemble averages.*

As an illustration, we simulate many random walks and calculate a histogram H_n of the particle position n at two different times. In the example below, we used

$$t = 20\Delta t \qquad \text{and} \qquad t = 40\Delta t.$$

We used a random-number generator which provides random numbers uniformly between 0 and 1. Such random number generators are standard for any programming language, and form the basis for all random number generations as we will see later. Whenever the random number R satisfies

$$R < \frac{1}{2},$$

we make a left jump, whereas if

$$R > \frac{1}{2},$$

we make a right jump.

We generated

$$N_{\text{ens}} = 2000$$

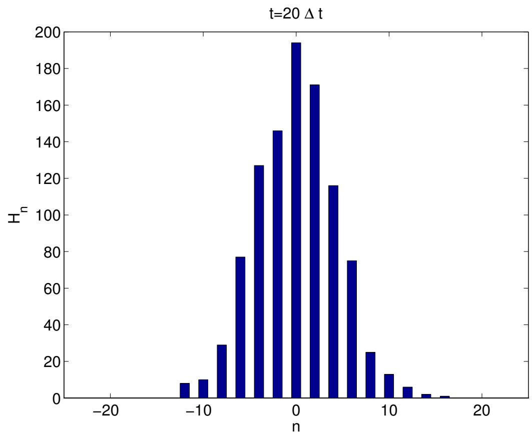
independent random walks, each starting at lattice site 0. For simplicity we set

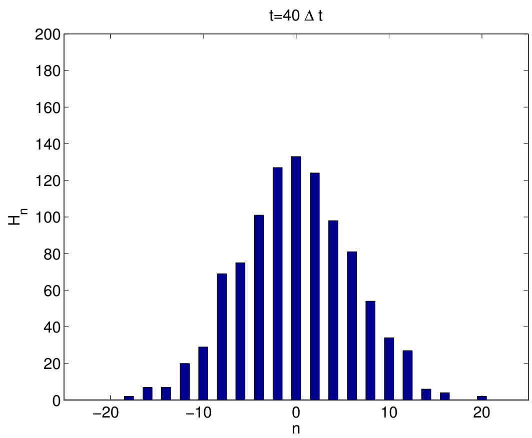
$$a = 1, \qquad \Delta t = 1.$$

The histogram H_n counts how many walkers are found at site n. To turn this into a probability distribution, we normalize the histogram:

$$p_n = \frac{H_n}{\sum_{n=-\infty}^{\infty} H_n}. \tag{1}$$

The quantity p_n is therefore the probability that the random walker is at lattice site n at time t.





Since the walker takes steps with equal probability to the left and right, the probability histogram is symmetric around $n = 0$. Notice also that n only takes even values in the figures above, because we used an even number of steps.

For longer times, the walker has had time to explore a larger region of space, and the distribution becomes wider. We will later see that for this simple random walk the variance grows linearly with time,

$$\langle x^2 \rangle \propto t,$$

while the typical width of the distribution grows like

$$\sqrt{\langle x^2 \rangle} \propto \sqrt{t}.$$

This square-root scaling is one of the characteristic signatures of diffusion.

The histograms also look approximately Gaussian. This is not a coincidence as we discuss a little bit later in this section.

2.2 Random variables

We now introduce the language of random variables.

A random variable X is a quantity whose value is not known in advance, but whose possible values occur with specified probabilities. For example, X may represent the outcome of throwing a die or a number obtained from a computer random-number generator.

For an ideal die, X can take the values

$$1, 2, 3, 4, 5, 6$$

with equal probability. For a random-number generator, X might instead be a real number between 0 and 1.

The point of introducing random variables is that they allow us to talk mathematically about uncertain outcomes. This is the basic language of stochastic processes, statistical physics, and many simulation methods.

2.3 Some definitions

A continuous random variable X is described by its probability density function $p(x) \geq 0$. The probability density is defined such that

$$p(x)dx = P\{x < X < x + dx\}, \tag{2}$$

where $P\{\cdots\}$ denotes the probability that the statement inside the curly brackets is true.

More generally,

$$\int_a^b p(x)\,dx = P\{a < X < b\}. \tag{3}$$

Since the random variable must take some value, the probability density must be normalized:

$$\int_{-\infty}^{\infty} p(x)\,dx = 1. \tag{4}$$

For discrete distributions, the integrals above are replaced by sums. An alternative is to use delta functions, as in the example below.

It is common practice to use upper-case letters for random variables, such as X, and lower-case letters for the corresponding argument of the probability density function, such as x.

Example: the die distribution. For an ideal die,

$$p_i = \frac{1}{6}, \qquad i = 1, 2, \ldots, 6.$$

In a continuum notation we can write this discrete distribution using Dirac delta functions:

$$p(x) = \sum_{i=1}^{6} p_i \delta(x - x_i). \tag{5}$$

Here $x_i = i$ and $\delta(z)$ is the Dirac delta function.

Example: the rectangular distribution. The rectangular, or uniform, distribution on the interval $a < x < b$ is

$$p(x) = \frac{1}{b-a}, \qquad a < x < b, \tag{6}$$

and $p(x) = 0$ otherwise.

This distribution gives equal probability density to all values between a and b. In the random-walk example above, we used a uniform random number with $a = 0$ and $b = 1$ to decide whether the walker should jump left or right.

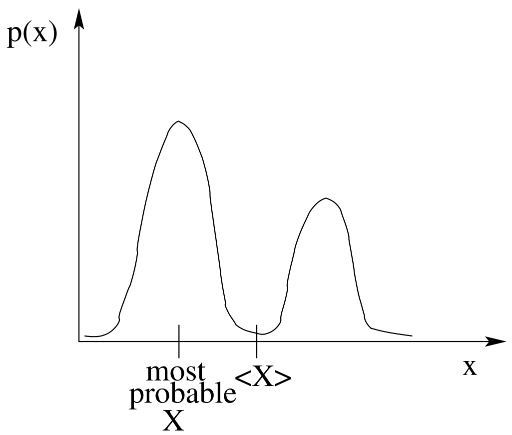
The average, or expectation value, of an arbitrary function $f(X)$ is

$$\langle f(X) \rangle = \int_{-\infty}^{\infty} f(x) p(x)\, dx. \tag{7}$$

Some standard averages are especially important. The *mean* is

$$\langle X \rangle = \int_{-\infty}^{\infty} x p(x)\, dx. \tag{8}$$

The mean is not necessarily the same as the most probable value, see the figure below.



The *variance* is

$$\sigma_X^2 = \left\langle (X - \langle X \rangle)^2 \right\rangle. \tag{9}$$

Expanding the square gives

$$\sigma_X^2 = \langle X^2 \rangle - 2\langle X \rangle^2 + \langle X \rangle^2,$$

and therefore

$$\sigma_X^2 = \langle X^2 \rangle - \langle X \rangle^2. \tag{10}$$

The variance measures the size of the fluctuations around the mean. The square root of the variance, σ_X is the *standard deviation*.

2.4 Transforming random numbers

On the computer, one typically has access to a pseudo-random-number generator that returns random numbers R uniformly distributed between 0 and 1:

$$p(r) = \begin{cases} 1, & 0 < r < 1, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Such random numbers are sometimes called rectangularly distributed.

But in simulations we often need random numbers from other distributions. For example, Brownian motion is naturally simulated using Gaussian random numbers, and waiting times in kinetic simulations are often exponentially distributed.

The question is therefore:

How can we transform uniform random numbers into random numbers with other distributions?

2.4.1 The transformation method — discrete case

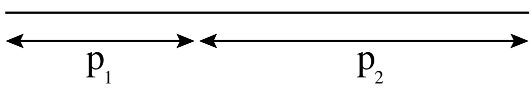
Suppose we want to generate a random number Y from a discrete probability distribution.

Consider first the case where Y takes two possible values, y_1 and y_2, with probabilities p_1 and p_2. Given a uniform random number R between 0 and 1, we choose y_1 if

$$R < p_1,$$

and y_2 if

$$R \geq p_1.$$



For the random walk above, we used precisely this method with

$$p_1 = p_2 = \frac{1}{2}.$$

For the general case, suppose that Y can take the values

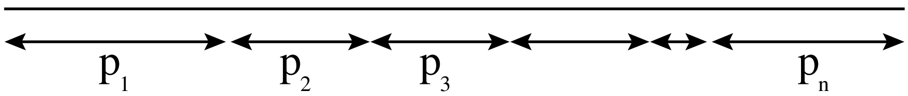
$$y_1, y_2, \ldots, y_n$$

with probabilities

$$p_1, p_2, \ldots, p_n.$$

We draw a uniform random number R and assign the value y_i to Y if

$$\sum_{j=0}^{i-1} p_j \leq R < \sum_{j=0}^{i} p_j.$$



It is convenient to introduce the cumulative distribution

$$C_i = \sum_{j=0}^{i} p_j, \tag{12}$$

where we define

$$p_0 \equiv 0.$$

The normalization condition becomes

$$C_n = 1.$$

The criterion above can then be written

$$C_{i-1} \leq R < C_i. \tag{13}$$

The procedure is therefore:

1. calculate the cumulative probabilities C_i,
2. draw a uniform random number R,
3. find the interval $C_{i-1} \leq R < C_i$,
4. assign the value $Y = y_i$.

This is the discrete transformation method. It is the basic idea behind many simple stochastic simulations.

2.4.2 The transformation method — continuum case

Suppose instead that we want to generate random numbers Y from a continuous probability density $p(y)$.

We introduce the cumulative distribution

$$C(y) = \int_{-\infty}^{y} p(y')\, dy'. \tag{14}$$

The quantity $C(y)$ is the probability that the random variable is smaller than y:

$$C(y) = P\{Y < y\}.$$

The transformation method says that if R is uniformly distributed between 0 and 1, then we choose Y such that

$$R = C(Y).$$

Equivalently,

$$Y = C^{-1}(R). \tag{15}$$

This is the continuous transformation method. The normalization condition

$$\int_{-\infty}^{\infty} p(y)\, dy = 1$$

is equivalent to

$$C(\infty) = 1.$$

Example: the exponential distribution. Suppose we have access to random numbers R uniformly distributed between 0 and 1, and we want random numbers with the distribution

$$p(y) = \begin{cases} \lambda e^{-\lambda y}, & y \geq 0, \\ 0, & y < 0, \end{cases} \qquad (\lambda > 0).$$

The first step is to calculate the cumulative distribution:

$$C(y) = \int_{-\infty}^{y} p(y')\, dy' = \int_{0}^{y} \lambda e^{-\lambda y'}\, dy' = 1 - e^{-\lambda y}.$$

The transformation $R \to Y$ is obtained by solving

$$C(Y) = R.$$

Thus,

$$1 - e^{-\lambda Y} = R,$$

or

$$Y = -\frac{1}{\lambda} \ln(1 - R).$$

Since $1 - R$ is also uniformly distributed between 0 and 1, we often write the result simply as

$$Y = -\frac{1}{\lambda} \ln R. \tag{16}$$

This formula is extremely useful. For example, in Gillespie simulations (see the sections on Molecular motors and one the Rate equations and dynamics in the cell), exponentially distributed waiting times appear naturally.

2.5 The normal distribution

The *normal distribution*, or Gaussian distribution, appears all over biophysics.

Why?

One reason is the central limit theorem. Roughly speaking, the theorem states that the sum of many independent random variables tends to become normally distributed, even if the individual random variables themselves are not normally distributed. Check out appendix A to learn more about this theorem. This is why the random-walk histograms at the beginning of this section look Gaussian. The final position of the walker is a sum of many independent steps.

The normal distribution with mean μ and variance σ^2 is defined through

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{17}$$

It is straightforward to show that

$$\langle X \rangle = \mu,$$

and

$$\sigma_X^2 = \sigma^2.$$

In simulations, Gaussian random numbers appear whenever we simulate Brownian motion in the continuum limit. For example,

$$x_{n+1} = x_n + \sqrt{2D\Delta t}\, \eta_n,$$

where η_n is a normally distributed random number with mean zero and variance one. Performing simulations using the recursion relation above would give us an alternative way to simulate the drunk student from earlier. Here, the student does not take steps of equal size, but the length can vary. Which of the models describe the motion of more realistically, we leave to the reader to text out empirically.

In appendix A we describe a method for turning uniformly distribution random numbers into normally distributed numbers. In practice, one usually uses the built-in random-number routines of a programming language. For example:

```
numpy.random.randn()
```

in Python,

```
randn
```

in Matlab.

2.6 Joint distributions

The simultaneous probability density, $p(x, y)$ of two random variables X and Y is called the *joint* PDF. It is defined by

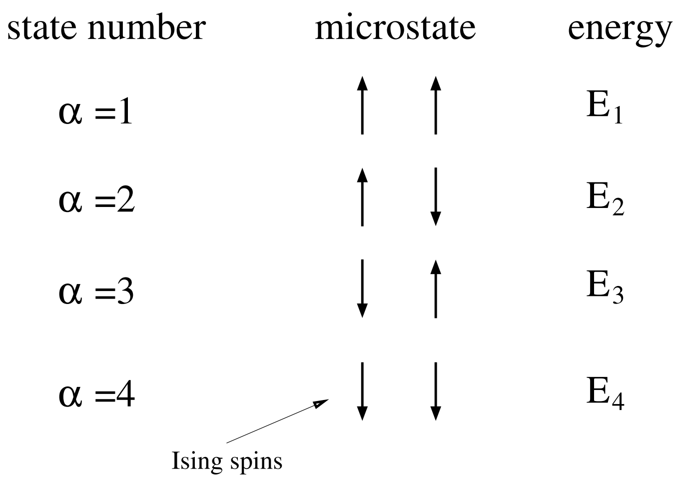
$$p(x, y)\, dx\, dy = P\{x < X < x + dx \text{ and } y < Y < y + dy\}. \tag{18}$$

2.7 The Boltzmann distribution

The Boltzmann distribution is one of the most important distributions in physics. It appears in statistical mechanics and biophysics. The random quantities X and Y can denote positions, conformational state of a biomolecule or arrangement of binding molecule onto a lattice such as DNA.

In statistical physics as applied to biological systems, we are in general considering a system in contact with a heat reservoir. The heat reservoir will have a fixed temperature $= T = 37^\circ$ C degrees in our bodies, or for in vivo experiment the T would correspond to room temperature.

We begin by identifying the possible microscopic states of a system. We label these states by an index α.



For example, the figure above depicts a simple system consisting of two spins, each of which can point in one of two directions,

$$s_i = \pm 1.$$

In a biophysics setting the states could represent bind/unbound states of two molecules (ligands) onto two adjacent sites on a DNA molecule.

The system has four possible states,

$$\boldsymbol{s} = (s_1, s_2).$$

We label these states by $\alpha = 1, 2, 3, 4$. For example, one state is

$$(s_1, s_2) = (1, 1),$$

another is

$$(s_1, s_2) = (-1, 1),$$

and so on. Associated with each state is an energy

$$E_\alpha.$$

Now suppose that the system is in contact with a heat bath at temperature T. Statistical mechanics says that, after sufficiently long times, the probability of finding the system in state α is

$$P_\alpha = \frac{\exp[-E_\alpha/(k_B T)]}{Z}, \tag{19}$$

where k_B is Boltzmann's constant. The normalization constant

$$Z = \sum_\alpha \exp[-E_\alpha/(k_B T)] \tag{20}$$

is called the partition function. The partition function is chosen so that

$$\sum_\alpha P_\alpha = 1.$$

Once we know the probabilities of the microstates, we can calculate the expectation value of any observable. If the value of an observable in state α is f_α, then

$$\langle f \rangle = \sum_\alpha f_\alpha P_\alpha. \tag{21}$$

This is the basic prescription of equilibrium statistical mechanics:

list the states, assign energies, compute probabilities, average.

Of course, the difficult part is often that the number of states is enormous.

Does the expression for P_α make physical sense? Let us check.

Consider again the two-spin system. Suppose that the states with parallel spins,

$$(1,1), \qquad (-1,-1),$$

have energy E_+, while the states with antiparallel spins,

$$(-1,1), \qquad (1,-1),$$

have energy E_-. Assume that

$$E_+ > E_-.$$

The probability of finding the system in one specific parallel state, say state 1, is

$$P_1 = \frac{\exp[-E_+/(k_B T)]}{2\exp[-E_+/(k_B T)] + 2\exp[-E_-/(k_B T)]}.$$

Dividing numerator and denominator by $\exp[-E_-/(k_B T)]$, we obtain

$$P_1 = \frac{\exp[-\Delta E/(k_B T)]}{2 + 2\exp[-\Delta E/(k_B T)]}, \tag{22}$$

where

$$\Delta E = E_+ - E_-.$$

If the energy difference is large compared with the thermal energy,

$$\Delta E \gg k_B T,$$

then

$$\exp[-\Delta E/(k_B T)] \ll 1,$$

and therefore

$$P_1 \approx 0.$$

The high-energy state is rarely occupied.

On the other hand, if

$$\Delta E = 0,$$

then all four states have the same energy, and

$$P_1 = \frac{1}{4}.$$

The system then spends equal probability in all four states. This is exactly what we would expect.

Finally, if the state space is continuous rather than discrete, the sums above are replaced by integrals. For example, for particles with continuous positions, a microstate may be specified by coordinates

$$\mathbf{x} = (x_1, x_2, \ldots, x_N),$$

and the Boltzmann distribution becomes a probability density function proportional to

$$e^{-E(\mathbf{x})/(k_B T)}.$$

This connection between energy, probability, and thermal fluctuations is one of the central pillars of biophysics.

More philosophically: We will often estimate whether the "typical energies", E_α, are larger than, comparable to, or smaller than the thermal energy, $k_B T$. And, we will often find that $E_\alpha \approx k_B T$, which is in fact a signature of life: the regime $E_\alpha \ll k_B T$ corresponds to a solid ("dead matter"), whereas $E_\alpha \gg k_B T$ corresponds to a gas (again, "dead matter").

3 Diffusion

At microscopic length scales in cell biology, thermal fluctuations play a central role. The reason is that, like we concluded at the end of the previous section, life is characterized by k_BT being or the same order as energy differences. In the special case where $E_\alpha = 0$, we get an "infinite degeneracy" (all positions have the same energy), and small particles which are suspended in a liquid are continuously bombarded by surrounding water molecules, leading to highly irregular motion. This phenomenon is known as Brownian motion. We encountered 1d version of Brownian motion in the previous section.

3.1 Diffusion as a transport mechanism

But, if the motion is completely random, can it be used for something useful, or is it just a nuisance? Well, in fact, diffusion is one of the most important transport mechanisms in biology. Examples include proteins diffusing inside cells, signaling molecules spreading through tissue, ions moving through channels, and nutrients being transported over microscopic distances. The important point is not only that diffusion occurs, but that its efficiency depends very strongly on length scale.

As we shall see, diffusion is very efficient over short distances, but becomes painfully slow over larger length scales. This simple observation has profound biological consequences.

The diffusion law, or, why do we have blood vessels?

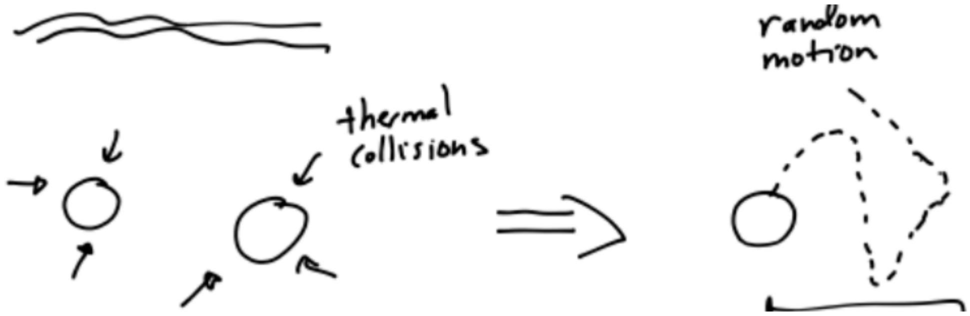


Figure 1: Thermal collisions leading to random motion.

Let us begin with a simple order-of-magnitude question:

How long does it take to travel a distance L by diffusion?

The characteristic diffusion time is

$$t = \frac{L^2}{D} \tag{23}$$

which we will refer to as the diffusion law. Here

$$[D] = \frac{\text{length}^2}{\text{time}}$$

and

$$D: \text{ diffusion constant.}$$

Eq. (23) will be derived more carefully later. For now, the important feature is the scaling

$$t \propto L^2.$$

This is very different from directed motion with a constant velocity, where $t \propto L$. In diffusion, doubling the distance does not double the time; it increases the time by a factor of four. Increasing the distance by a factor of 10^3 increases the time by a factor of 10^6.

Estimate: For a large globular protein, a typical value of the diffusion constant is

$$D \sim 10^{-10}\ \frac{\mathrm{m}^2}{\mathrm{s}}.$$

We will derive this estimate later, using the Einstein relation together with Stokes law.

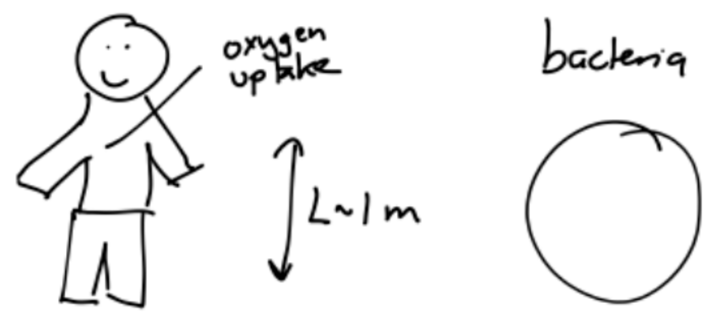


Figure 2: Diffusion law examples for humans and bacteria.

Let us first try the diffusion law on a human length scale.

Human:

$$L \sim 1\ \mathrm{m}.$$

Using Eq. (23),

$$t = \frac{L^2}{D} = \frac{1^2}{10^{-10}}\ \mathrm{s}$$

and therefore

$$t = 10^{10}\ \mathrm{s} \approx 300\ \text{years}.$$

Pure diffusion is therefore far too slow for transport across the human body. If nutrients, oxygen, or signaling molecules had to diffuse over meter distances, we would be in serious trouble.

The biological solution is directed transport. If material is transported with a typical velocity V, then

$$t = \frac{L}{V}.$$

This linear scaling with distance is enormously more efficient over large length scales. In multicellular organisms this is one of the physical reasons why we need blood vessels.

Now let us repeat the same estimate on a bacterial length scale.

Bacteria:

$$L \sim 10^{-6} \text{ m}.$$

Then

$$t = \frac{L^2}{D} = \frac{(10^{-6})^2}{10^{-10}} \text{ s} = 10^{-2} \text{ s} = 10 \text{ ms}.$$

Diffusion is therefore a perfectly reasonable transportation mechanism over bacterial length scales. A molecule can explore a micron-sized cell on a biologically relevant time scale.

This simple estimate already illustrates how strongly biological organization depends on physical length scales. The same physical process which is useless over a meter can be extremely efficient over a micron.

3.2 The diffusion equation

Let us now revisit the random walker on a lattice problem from the previous section. Our aim is to derive an equation for the PDF to be at position x at time t. This equation will then be, as we will see, be easy to extend to more dimensions, and in later sections we will extend this equation to include drift terms.

During each small time interval, the particle can jump either to the left or to the right.

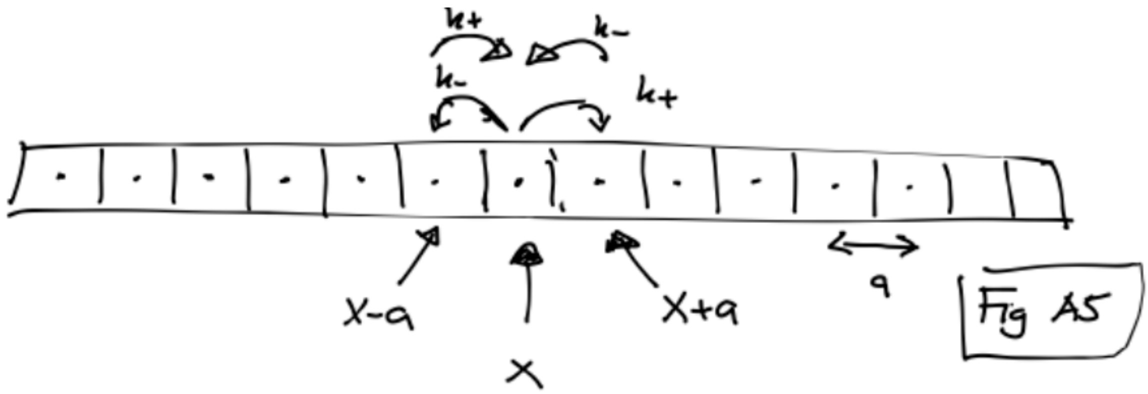


Figure 3: One-dimensional hopping model.

Let $p(x,t)$ denote the probability of finding the particle at position x at time t. We now write down a balance equation for the probability at site x.

The change in probability is

$$p(x, t + \Delta t) - p(x, t).$$

There are three contributions to this change.

First, the particle may jump into site x from the left, that is from $x - a$ to x. This gives the contribution

$$+\Delta t \, k_+ \, p(x - a, t),$$

where k_+ is the rate for jumping to the right.

Second, the particle may jump into site x from the right, that is from $x + a$ to x. This gives

$$+\Delta t \, k_- \, p(x + a, t),$$

where k_- is the rate for jumping to the left.

Finally, the particle may leave site x, either by jumping to the right or by jumping to the left. This gives the loss term

$$-\Delta t(k_+ + k_-)p(x,t).$$

Hence,

$$p(x,t+\Delta t) - p(x,t) = \Delta t\, k_+\, p(x-a,t) + \Delta t\, k_-\, p(x+a,t) - \Delta t(k_+ + k_-)p(x,t). \tag{24}$$

This equation is a master equation in discrete space. It is still exact for the simple hopping model. To obtain a continuum equation, we now assume that the lattice spacing a is small compared with the length scale over which $p(x,t)$ changes appreciably. And, we focus on the symmetric case

$$k_+ = k_- = k,$$

which corresponds to diffusion without drift.

Taylor expanding with respect to a gives

$$p(x+a,t) \approx p(x,t) + a\frac{\partial p(x,t)}{\partial x} + \frac{a^2}{2}\frac{\partial^2 p(x,t)}{\partial x^2} \tag{25}$$

and similarly

$$p(x-a,t) \approx p(x,t) - a\frac{\partial p(x,t)}{\partial x} + \frac{a^2}{2}\frac{\partial^2 p(x,t)}{\partial x^2}.$$

Divide Eq. (24) by Δt and insert the Taylor expansions. Then let $\Delta t \to 0$. We obtain

$$\frac{\partial p(x,t)}{\partial t} \approx 2kp(x,t) + a(-k+k)\frac{\partial p(x,t)}{\partial x}$$

$$+\frac{a^2}{2}(2k)\frac{\partial^2 p(x,t)}{\partial x^2} - (2k)p(x,t).$$

The zeroth-order terms cancel. The first-derivative term also cancels because the walk is symmetric. We are left with

$$\frac{\partial p(x,t)}{\partial t} = D\frac{\partial^2 p(x,t)}{\partial x^2}, \tag{26}$$

where

$$D = \frac{a^2(k_+ + k_-)}{2}. \tag{27}$$

For the symmetric case $k_+ = k_- = k$, this becomes simply

$$D = a^2 k.$$

Eq. (26) is the diffusion equation.

This equation is one of the most important equations in non-equilibrium statistical physics. Individual trajectories are highly irregular, but the probability density evolves in a smooth and predictable way. Closely related equations appear in many different areas of physics, including heat conduction, particle transport, chemical spreading, and intracellular signaling.

It is very instructive to simulate random walks numerically. Even a few lines of code are sufficient to reproduce many of the analytical results derived below, see the previous section. One may perform the simulation on a lattice, or in the continuum.

The solution to the diffusion equation

$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2}$$

for a particle initially located at $x = x_0$ is

$$p(x,t) = \frac{\exp\left[-\frac{(x-x_0)^2}{4Dt}\right]}{\sqrt{4\pi Dt}}. \tag{28}$$

where we assume an initial condition $p(x, t = 0) = \delta(x - x_0)$, where $\delta(x - x_0)$ is the Dirac delta function.

The Gaussian form of Eq. (28) reflects the central limit theorem: many independent random steps combine to produce a normal distribution. The width of the distribution grows as

$$\sqrt{Dt},$$

meaning that the probability distribution continuously broadens with time.

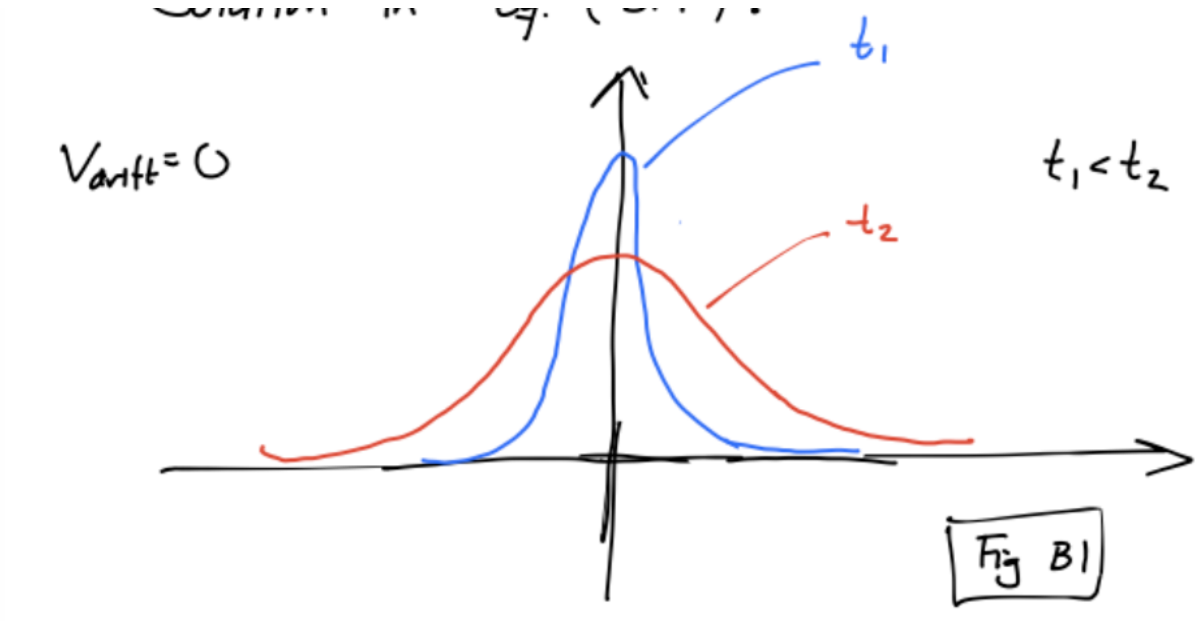


Figure 4: Solution in Eq. (28) for different times.

We now calculate a few averages. Denote by

$$\langle \cdots \rangle = \text{ensemble average.}$$

In an experiment this would mean repeating the same experiment many times. In a simulation it means generating many independent trajectories and then averaging over them.

For simplicity, let us first take $x_0 = 0$. Then the distribution is symmetric around the origin, and therefore

$$\langle (x - x_0) \rangle = 0.$$

This does not mean that the particle does not move. It means that positive and negative displacements are equally likely. A better measure of the distance traveled is the mean-squared displacement.

Using Eq. (28),

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 \frac{\exp\left[-\frac{x^2}{4Dt}\right]}{\sqrt{4\pi Dt}}\, dx = 2Dt. \tag{29}$$

More generally, for a particle starting at x_0,

$$\langle (x - x_0)^2 \rangle = 2Dt.$$

This is the precise mathematical statement of the diffusion law in one dimension. The characteristic distance traveled by diffusion therefore scales as

$$x_{\text{typical}} \sim \sqrt{Dt}.$$

Equivalently, to travel a distance L by diffusion requires a time of order

$$t \sim \frac{L^2}{D},$$

which is the estimate we used at the beginning of this section. Numerical factors such as 2 depend on the precise definition of "typical distance", but the important physical scaling is

$$t \propto L^2.$$

This should be contrasted with ordinary ballistic motion, where distance grows linearly with time.

A useful numerical exercise is to generate many random walks and verify that the mean-squared displacement grows linearly with time. A typical workflow is:

```
for each trajectory:
    simulate x(t)
    store (x(t)-x(0))**2

average over trajectories
plot MSD versus time
```

The slope of the MSD curve gives $2D$ in one dimension. This is one of the simplest ways of estimating a diffusion constant from simulated or experimental trajectories.

3.3 First approach: How to estimate the diffusion constant from trajectories

So far, we have treated the diffusion constant D as a known physical parameter. In experiments and simulations, however, D is often precisely what we want to estimate.

This is a very common situation in biophysics. We observe a stochastic trajectory,

$$x_0, x_1, x_2, \ldots, x_N,$$

sampled at times

$$t_i = i\Delta t,$$

and we would like to infer the diffusion constant from the data.

Simplest way of estimating D is to use the mean-squared displacement. From the diffusion law in one dimension,

$$\langle (\Delta x)^2 \rangle = 2D\Delta t.$$

Therefore, if we have many observed displacements, a natural estimator is

$$\hat{D}_{\text{MSD}} = \frac{1}{2\Delta t}\frac{1}{N}\sum_{i=0}^{N-1}(\Delta x_i)^2. \tag{30}$$

Equivalently,

$$\hat{D}_{\text{MSD}} = \frac{1}{2N\Delta t}\sum_{i=0}^{N-1}(x_{i+1} - x_i)^2. \tag{31}$$

This estimator is easy to understand physically. We measure how large the typical squared displacement is during one time step, and then divide by $2\Delta t$.

A tiny numerical version of the idea is:

```
dx = x[1:] - x[:-1]
Dhat = np.mean(dx**2)/(2*dt)
```

In Matlab this would look almost the same, using `diff(x)`.

The estimator in Eq. (31) is simple, intuitive, and often a good first thing to try. But it is also useful to view the same problem from a more systematic statistical perspective.

3.4 Second approach: How to estimate the diffusion constant from trajectories – Bayes theorem

Now, we introduce a more powerful approach, Bayesian data analysis, for estimating the diffusion constant. We will see this approach will lead to similar final estimates, but has the strength that it can be applied to any type of stochastic model with arbitrary number of model parameters.

Suppose that we observe the full set of displacements

$$\Delta x_0, \Delta x_1, \ldots, \Delta x_{N-1}.$$

In the Bayesian setting this set is referred to as the "data".

We now seek to introduce a model describing the data. If the trajectory is Brownian, then the increments are independent. Therefore, the probability density of observing the whole trajectory, given D, is the product of the single-step probabilities:

$$L(D) = p(\Delta x_0, \Delta x_1, \ldots, \Delta x_{N-1}|D) = \prod_{i=0}^{N-1} p(\Delta x_i|D). \tag{32}$$

Using the solution of the diffusion equation, Eq. (28), for each "time slice", we get

$$L(D) = \prod_{i=0}^{N-1} \frac{1}{\sqrt{4\pi D\Delta t}} \exp\left[-\frac{(\Delta x_i)^2}{4D\Delta t}\right]. \tag{33}$$

Combining the factors gives

$$L(D) = (4\pi D\Delta t)^{-N/2} \exp\left[-\frac{1}{4D\Delta t}\sum_{i=0}^{N-1}(\Delta x_i)^2\right]. \tag{34}$$

It is usually more convenient to work with the logarithm of the likelihood:

$$\log L(D) = -\frac{N}{2}\log(4\pi D\Delta t) - \frac{1}{4D\Delta t}\sum_{i=0}^{N-1}(\Delta x_i)^2. \tag{35}$$

The maximum-likelihood estimate is the value of D that maximizes $L(D)$, or equivalently $\log L(D)$. Let

$$S = \sum_{i=0}^{N-1}(\Delta x_i)^2.$$

Then, ignoring terms independent of D,

$$\log L(D) = -\frac{N}{2}\log D - \frac{S}{4D\Delta t} + \text{constant}.$$

Differentiating with respect to D gives

$$\frac{d}{dD}\log L(D) = -\frac{N}{2D} + \frac{S}{4\Delta t}\frac{1}{D^2}.$$

Setting this derivative equal to zero,

$$-\frac{N}{2D} + \frac{S}{4\Delta t}\frac{1}{D^2} = 0.$$

Multiplying by D^2 gives

$$-\frac{N}{2}D + \frac{S}{4\Delta t} = 0,$$

and therefore

$$\hat{D}_{\mathrm{ML}} = \frac{S}{2N\Delta t} = \frac{1}{2N\Delta t}\sum_{i=0}^{N-1}(\Delta x_i)^2. \tag{36}$$

Thus, in this simple case, the maximum-likelihood estimator is identical to the single-time-step

MSD estimator.

This is a nice result. It shows that the intuitive MSD estimate is not just a reasonable guess; for ideal Brownian motion with independent Gaussian increments, it is also the maximum-likelihood estimate.

But, the maximum likelihood estimate (or MSD fit) does not tell us anything about how good the estimate is. In fact, we can be dead certain that is is not spot-on correct! So, we need also a "error" on our estimate for it to make sense. Enter: the Bayes theorem.

The likelihood tells us the probability of observing the data for a given value of D:

$$p(\text{data}|D).$$

But in many situations we would like to turn the question around. Given the observed data, what can we say about the possible values of D?

This is where Bayes' theorem enters.

For two events or hypotheses A and B, Bayes' theorem is

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}. \tag{37}$$

In parameter inference, the same formula becomes

$$p(D|\text{data}) = \frac{p(\text{data}|D)p(D)}{p(\text{data})}. \tag{38}$$

Each factor has a name:

$$p(D|\text{data}) = \text{posterior},$$

$$p(\text{data}|D) = \text{likelihood},$$

$$p(D) = \text{prior},$$

and

$$p(\text{data}) = \text{evidence}.$$

The evidence is given by

$$p(\text{data}) = \int_0^\infty p(\text{data}|D)p(D)\, dD. \tag{39}$$

For estimating a diffusion constant, the posterior distribution is

$$p(D|\Delta x_0, \ldots, \Delta x_{N-1}) = \frac{L(D)p(D)}{\int_0^\infty L(D)p(D)\, dD}. \tag{40}$$

This expression contains the central idea of Bayesian inference. We begin with some prior knowledge about plausible values of D, encoded in $p(D)$. We then update this knowledge using the measured trajectory, through the likelihood $L(D)$. The result is the posterior distribution.

The posterior is not just a single best estimate. It is a probability distribution over possible values of D. This is useful because it also quantifies uncertainty, so it automatically give as an "error" for any estimate.

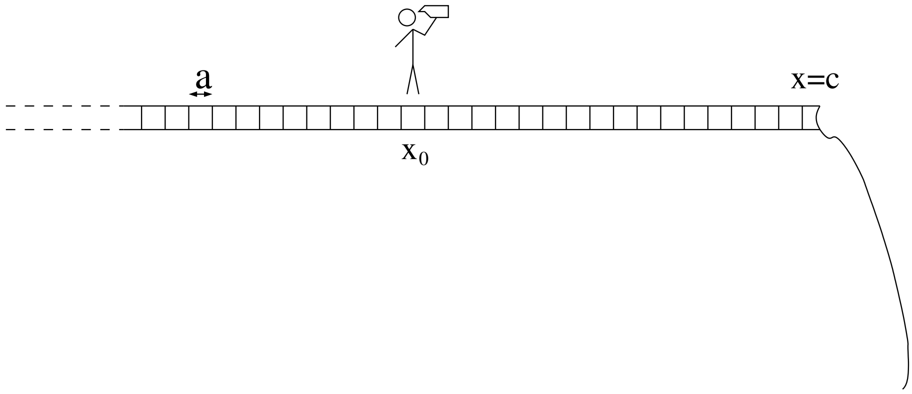
A simple computational workflow is:

```
D_values = np.linspace(Dmin, Dmax, 1000)

for each D in D_values:
    compute log_likelihood(D)

posterior = exp(log_likelihood) * prior
posterior = posterior / integral(posterior over D)
```

In actual code, it is often better to work with the log-likelihood to avoid numerical underflow, since products of many small probabilities can become extremely small.

3.5 First-passage time problems

Let us now turn back to the problem introduced in the first section – a random walker on a one-dimensional line. To spice things up a little we further assume that we position our random walker near a cliff located at $x = c$, see figure below. Or, if you prefer less drama, we assume the absorbing point is the dorm room of a drunk student.



The natural question to ask is then: how long will it take before the random walker falls off the cliff (finds her/his dorm room)? Since the random walk is a stochastic process different simulations will give different results, and we are interested in the distribution $f(t)$ of times it takes to reach the cliff. This is the simplest example of a *first-passage time problem.* It is easy to generalize the problem above to higher dimensions and and to random walkers in a potential etc; such systems can straightforwardly be simulated in analogous fashions to what we do below. The present problem has the advantage that it can be solved analytically, so let us stick with the simple one-dimensional, potential-free problem for now. More exotic first-passage time problems are discussed at length in the book *Sidney Redner, A guide to first-passage time processes.*

Consider a random walker starting at position x_0 with an absorbing barrier at $x = c$ ($x_0 < c$). To left of the random walker we assume that the system extends to infinity. If we assume that the absorbing point c is located many lattice sites away from x_0 the random walker will need many steps before reaching c. The process will then be described as a sum of many independent identically distributed random numbers and the probability density function will be described, for $x < c$, by the diffusion equation, Eq. (125), i.e.

$$\frac{\partial p(x,t|x_0)}{\partial t} = D\frac{\partial^2 p(x,t|x_0)}{\partial x^2}, \tag{41}$$

with initial condition $p(x,t|x_0) = \delta(x - x_0)$. At $x = c$ we introduce the absorbing boundary condition:

$$p(x = c, t|x_0) = 0. \tag{42}$$

The solution to the equations above is

$$p(x,t|x_0) = \psi(x - x_0) - \psi(x - (2c - x_0)), \tag{43}$$

where

$$\psi(z) = \frac{1}{\sqrt{4\pi Dt}}\mathrm{e}^{-\frac{z^2}{4Dt}}. \tag{44}$$

It is straightforward to check that indeed the solution above satisfies Eqs. (41) and (42).

We now seek the first-passage time density $f(t)$. The (survival) probability that the particle is still at a position $x < c$ at time t is

$$S(t) = \int_{-\infty}^{c} p(x,t|x_0). \tag{45}$$

The probability that walker fell off the cliff at times between t and $t+dt$ is $f(t)dt = S(t) - S(t+dt)$, i.e. the first-passage time density is:

$$f(t) = -\frac{\partial S(t)}{\partial t} \tag{46}$$

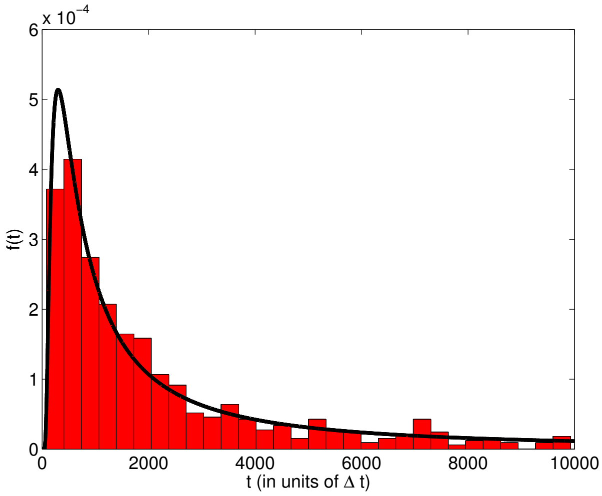
If we plug in Eq. (43) into Eq. (45) we find (after a little bit of work) the survival probability

$$S(t) = \frac{2}{\sqrt{\pi}}\int_0^{(c-x_0)/\sqrt{4Dt}} e^{-s^2}ds = \mathrm{erf}(\frac{c - x_0}{\sqrt{4Dt}}). \tag{47}$$

The first-passage time density then becomes:

$$f(t) = \frac{c - x_0}{\sqrt{4\pi Dt^3}}\mathrm{e}^{-\frac{(c-x_0)^2}{4Dt}}. \tag{48}$$

For long times we thus have $f(t) \sim t^{-3/2}$. In the figure below we show a comparison between the expression above and simulation results using the same approach as in the first section. In the present case whenever the walker reached the lattice site c we terminated the simulation and stored the corresponding first-passage time. This procedure was repeated $N_{\mathrm{ens}} = 1000$ times.



We used $x_0 = 0$ and $c = 30$ in the simulation, and the data was binned into 30 bins. The bars show the results of the simulation and the solid curve is the analytic result. Note that in first-passage simulations of the present type we need to set a stop-time for the simulation, to avoid "unsuccessful" events where the random walkers strays off towards minus infinity and never reach c. When normalizing the histograms these unsuccessful events must be included. In the simulation results above we set a stop-time equal to $10^4 \Delta t$, which, in this particular random number realization, gave 230 "unsuccessful" events.

3.6 Derivation of the solution of the diffusion equation using a path integral approach*

The result Eq. (28) can also be obtained using a path integral approach. This is closely related in spirit to the path integral formulation of quantum mechanics due to Feynman.

The central idea is that there are many possible stochastic trajectories connecting two points. The probability distribution is obtained by summing over all such paths.

$$\text{diffusion equation} \Leftrightarrow \text{Schrödinger equation}$$

(no drift) if we formally replace

$$t \to it.$$

This connection is not needed for practical simulations of diffusion, but it is a beautiful example of how the same mathematical structures appear in different areas of physics.

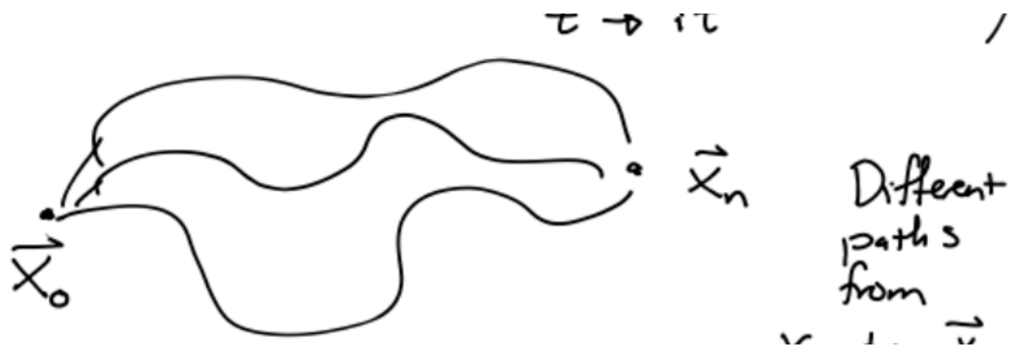


Figure 5: Different paths from x_0 to x_n.

Let us discretize time into N small intervals of length Δt, so that

$$t = N\Delta t.$$

A path is then specified by the intermediate positions

$$x_1, x_2, \ldots, x_{N-1},$$

with fixed endpoints x_0 and $x_N = x$. For Brownian motion, each small step is Gaussian distributed:

$$p(x_{i+1}|x_i) = \frac{1}{\sqrt{4\pi D\Delta t}} \exp\left[-\frac{(x_{i+1}-x_i)^2}{4D\Delta t}\right].$$

The probability for one particular discretized path is therefore the product

$$\prod_{i=0}^{N-1} \frac{1}{\sqrt{4\pi D\Delta t}} \exp\left[-\frac{(x_{i+1}-x_i)^2}{4D\Delta t}\right].$$

Combining the exponentials gives

$$\prod_{i=0}^{N-1} p(x_{i+1}|x_i) = (4\pi D\Delta t)^{-N/2} \exp\left[-\sum_{i=0}^{N-1} \frac{(x_{i+1}-x_i)^2}{4D\Delta t}\right].$$

To obtain the total probability of going from x_0 to x in time t, we integrate over all intermediate positions:

$$p(x,t|x_0,0) = \int dx_1 \cdots dx_{N-1} \prod_{i=0}^{N-1} p(x_{i+1}|x_i).$$

In the continuum limit this becomes a sum over all paths,

$$p(x,t|x_0,0) = \int_{x(0)=x_0}^{x(t)=x} \mathcal{D}x(\tau)\, \exp\left[-\int_0^t \frac{\dot{x}(\tau)^2}{4D}\, d\tau\right],$$

up to a normalization factor.

This expression says something very physical. All paths are allowed, but paths with very large rapid changes have a large value of $\dot{x}^2$ and are therefore exponentially suppressed. The most probable path is the straight path, but the probability distribution is obtained by including fluctuations around it.

Carrying out the Gaussian path integral gives back the diffusion kernel

$$p(x,t|x_0,0) = \frac{\exp\left[-\frac{(x-x_0)^2}{4Dt}\right]}{\sqrt{4\pi Dt}},$$

which is Eq. (28).

This derivation is perhaps more advanced than what is needed for the rest of the course, but it is useful because it connects diffusion, stochastic processes, statistical physics, and quantum mechanics in one common mathematical language.

4 The mathematics of water

So far, the role of water has mostly been to provide random thermal kicks to molecules. Water was the invisible background responsible for Brownian motion. In this section we look more closely at the water itself.

This is important because the microscopic world is not only noisy; it is also very viscous. At cellular length scales, motion through water is completely different from motion through air at human length scales. If we stop pushing a macroscopic object, it may keep moving for a while because of inertia. If a bacterium stops swimming, it essentially stops immediately. The world of cells is a world where friction dominates.

4.1 The Navier-Stokes equation – in five minutes

The mathematical description of fluid motion is given by the Navier–Stokes equation. For an incompressible Newtonian fluid, it reads

$$\rho \left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right] = -\nabla p + \eta \nabla^2 \mathbf{v}. \tag{49}$$

This equation is supplemented by the incompressibility condition

$$\nabla \cdot \mathbf{v} = 0.$$

Here

$$\rho = \text{mass density}, \qquad \eta = \text{dynamic viscosity},$$

$$\mathbf{v}(\mathbf{r}, t) = \text{fluid velocity}, \qquad p(\mathbf{r}, t) = \text{pressure}.$$

The Navier–Stokes equation may look complicated, but is essentially Newton's second law for a small fluid element. Derivation of the Navier-Stokes equations and the incompressible fluid in found in appendix B. The left-hand side is mass density times acceleration. The first term,

$$\rho \frac{\partial \mathbf{v}}{\partial t},$$

is the local acceleration of the fluid. The second term,

$$\rho(\mathbf{v} \cdot \nabla)\mathbf{v},$$

is the convective acceleration, which appears because a fluid element moves from one place to another where the velocity field may be different.

The right-hand side contains the forces per unit volume. The pressure gradient term,

$$-\nabla p,$$

pushes fluid from regions of high pressure toward regions of low pressure. The viscous term,

$$\eta \nabla^2 \mathbf{v},$$

describes the tendency of viscosity to smooth out velocity differences. Viscosity can be though of as a measure of "how difficult is it to push an object through the liquid". Thus, water has a lower viscosity than honey, which in turn have a lower viscosity than asphalt. The viscosity of water is 10^{-3} Pa$\cdot$ s; this is a good value to memorize for when we later do order of magnitude estimates.

Together with incompressibility, the equations above form a closed set of four equations for the four unknown fields

$$v_x, \quad v_y, \quad v_z, \quad p.$$

For now, we will use the equation as a tool for understanding some simple but biologically important flows.

The standard boundary condition between a liquid and a solid surface is the no-slip boundary condition. This means that the fluid velocity at the surface is equal to the velocity of the surface itself. There is also no penetration through the solid boundary. Thus, if the wall is stationary, the fluid velocity at the wall is zero. This boundary condition may look innocent, but it has important consequences. It is one of the reasons why thin tubes have large flow resistance and why microscopic swimming is so different from macroscopic swimming.

4.2 Flow through tubes — Poiseuille flow

An important biological example is fluid flow through cylindrical tubes, for example blood flow through blood vessels.

We consider stationary flow through a cylindrical tube of radius R and length L, see Fig. 6. The pressure difference between the two ends of the tube is Δp. In appendix C, we give derive the liquid flow profile in the channel, assuming that the flow is stationary.

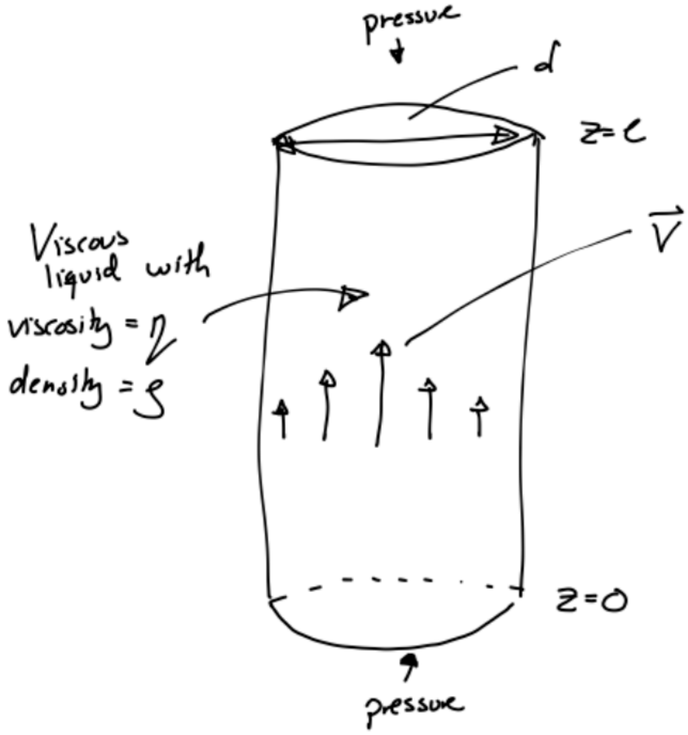


Figure 6: Tube geometry for Poiseuille flow.

the result is a flow profile:

$$v_z(r) = \frac{\Delta p}{4\eta L}\left(R^2 - r^2\right). \tag{50}$$

The velocity profile is therefore parabolic. The fluid moves fastest in the center of the tube and is zero at the wall due to the boundary condition. The flow is inversely proportional to the viscosity – it is probably not as surprise that a more viscous liquid with move slower through a tube.

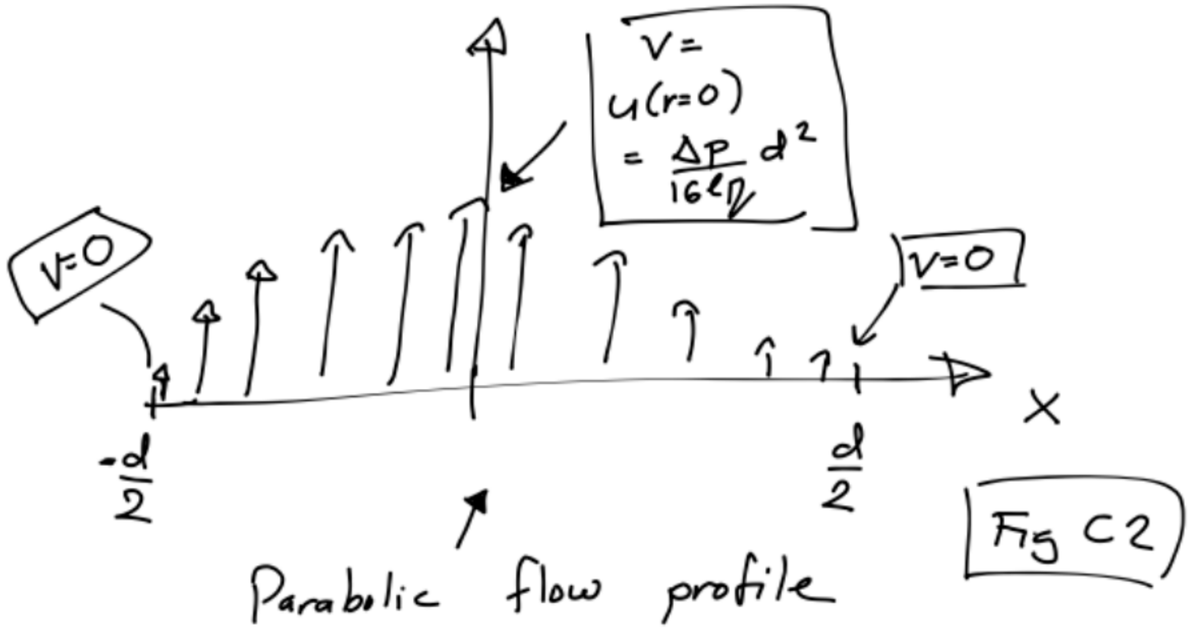


Figure 7: The Poiseuille velocity profile.

This is a nice example of how geometry, viscosity, and biological function are tied together by a simple physical law.

4.3 Low Reynolds numbers and Stokes law

In the previous subsection we were lucky: by pure symmetry the non-linear term in the Navier-Stokes equations were $= 0$, and the equations could be solved. In general, this is not the case, however. But, in biophysics, luck strikes again: the non-linear term can often be neglected because of the smallness of a famous dimensionless number called *The Reynolds number*.

We now ask a basic physics question: which terms in the Navier–Stokes equation are important? The answer depends on length scale, velocity, density, and viscosity. The dimensionless number which measures this is the Reynolds number.

We start from the Navier–Stokes equation,

$$\rho\frac{\partial \mathbf{V}}{\partial t} + \rho(\mathbf{V}\cdot\nabla)\mathbf{V} = -\nabla p + \eta\nabla^2\mathbf{V}. \tag{51}$$

Let us denote the two inertial terms by

$$A \equiv \rho\frac{\partial \mathbf{V}}{\partial t},$$

and

$$B \equiv \rho(\mathbf{V}\cdot\nabla)\mathbf{V}.$$

The viscous term is

$$C \equiv \eta\nabla^2\mathbf{V}.$$

The question is:

Which terms are dominant, and when?

To answer this, we use an order-of-magnitude argument. Such arguments are common in biophysics. They allow us to do useful physics without solving the full equations.

Consider a body of typical length L moving with velocity U.

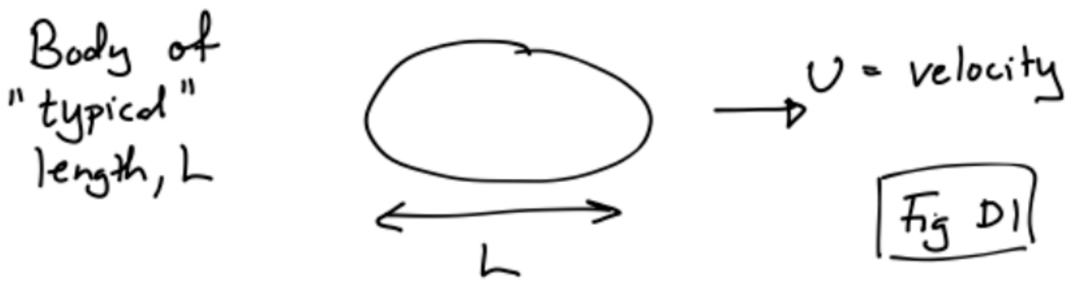


Figure 8: Characteristic object of size L moving with velocity U.

We estimate the local acceleration term as

$$A \approx \rho \frac{U}{\Delta t}.$$

The natural time scale for the flow to change over a distance L is

$$\Delta t = \frac{L}{U}.$$

Therefore,

$$A \approx \rho \frac{U^2}{L}.$$

The convective acceleration has the same order of magnitude:

$$B \approx \rho \frac{U^2}{L}.$$

For the viscous term, each spatial derivative contributes roughly a factor $1/L$, and therefore

$$C \approx \eta \frac{U}{L^2}.$$

The ratio between inertial and viscous effects is therefore

$$\frac{A+B}{C} \sim \frac{\rho U^2/L}{\eta U/L^2} = \frac{\rho U L}{\eta}.$$

This dimensionless quantity is called the Reynolds number:

$$\mathrm{Re} = \frac{\rho U L}{\eta}. \tag{52}$$

Let us consider a bacterium swimming at a typical velocity $U = 10\ \mu$m, of size $L = 1\ \mu$m in

water, we find
$$\mathrm{Re} \sim \frac{10^3 \cdot 10^{-6} \cdot 10^{-5}}{10^{-3}} = 10^{-5}.$$
Indeed, we find that the Reynolds number is small and we can neglect inertia. We used that the density of water is $\rho = 1000$ kg/m^3, and our memorized value for the viscosity of water (right, you remember...?). For a globular protein, the Reynolds number is even smaller.

As a contrast, consider a fish swimming at a velocity $U = 1$ m/s, and of size $L = 0.1$ m. We then have:
$$\mathrm{Re} \sim \frac{10^3 \cdot 0.1 \cdot 1}{10^{-3}} = 10^5.$$
where inertia dominates viscosity effect dramatically.

Thus, on cellular and molecular length scales, viscous forces dominate over inertial forces. In practice, this means that the left-hand side of Eq. (51) can often be neglected, and the fluid equation becomes
$$0 = -\nabla p + \eta \nabla^2 \mathbf{V}.$$
This is the equation we almost always start out with when working with biophysics at the cellular length scale. Note that the only physical parameter which enters is the viscosity.

4.4 Stokes law: Particle moving through a liquid at $\mathrm{Re} \ll 1$

An important property of any particle/molecule moving through a liquid is its friction coefficient, γ. It is defined as the constant of proportionality between the force applied to a particle and the mean velocity U of that particle:
$$F = \gamma U. \tag{53}$$
Large γ means that a large force is required to maintain a given velocity.

To derive an explicit expression for γ, one solves Eqs. (4.3) for the fluid velocity and pressure around a sphere of radius R. The sphere moves with velocity U relative to the surrounding fluid. At the surface of the sphere we apply the no-slip and no-penetration boundary conditions, and far away from the sphere the fluid velocity approaches the imposed background flow. The full calculation is a classic result in fluid mechanics. The velocity disturbance created by the sphere decays slowly with distance from the particle. At large distances r from the particle, the scale of the velocity disturbance behaves as
$$|\mathbf{v}(\mathbf{r})| \sim \frac{R}{r}.$$
This results has important consequences and in general leads to so called hydrodynamic couplings between particle/molecules/surfaces. We will not dwell further into this, but it is worth remembering if one want to solve a problem involving several particles, these type of fluid-induced coupling cannot be neglected.

Moving on with our single particle, the friction coefficient is obtained by evaluating the local viscous and pressure forces on the sphere and integrating these over the surface. The result is
$$\gamma = 6\pi\eta R. \tag{54}$$

This result is the famous *Stokes law.* We see that a larger particle has a larger friction coefficient, and that a more viscous liquid also gives a larger friction coefficient.

Before accepting the result, let us see what *dimensional analysis* can tell us. This is often a very useful trick in biophysics. We seek a relation between the force and the velocity. Since the equations are linear we must have

$$F = \gamma U.$$

Since

$$[F] = \mathrm{kg\,m\,s^{-2}}, \qquad [U] = \mathrm{m\,s^{-1}},$$

we have that

$$[\gamma] = \mathrm{kg\,s^{-1}}.$$

What quantities can go into γ? Well, at low Reynolds number we only have η and the radius of the sphere, R. No other quantities appear in the equations. The viscosity has units

$$[\eta] = \mathrm{Pa\,s} = \frac{\mathrm{kg}}{\mathrm{m\,s}},$$

and the particle radius has units

$$[R] = \mathrm{m}.$$

Staring at these dimensions for a while, we see that the only combination of R and η which has units of $\mathrm{kg\,s^{-1}}$ is

$$[\eta R] = \frac{\mathrm{kg}}{\mathrm{m\,s}}\mathrm{m} = \mathrm{kg\,s^{-1}}.$$

So dimensional analysis predicts

$$\gamma \sim \eta R.$$

This is the only combination of the parameters of the problem which ends up with the right dimension in the low Reynolds number regime. Had we not been in this regime we would have had another length scale set by the particle density and life had been a bit more difficult. Note that our dimensional analysis does not give the numerical prefactor, but it does give the correct dependence on viscosity and particle size.

This is a nice example of what dimensional analysis can and cannot do. It can often give the correct scaling almost for free, but in general not the exact numerical value for the prefactor.

And, in fact: Always check units! Dimensional analysis-type reasoning is useful also if you managed to derive an analytical prediction: checking units are often the fastest way to catch a wrong equation or to understand what a parameter means physically.

4.5 Harmonic spring motion in a viscous liquid

Consider a small particle attached to a harmonic spring and immersed in a liquid with viscosity η.

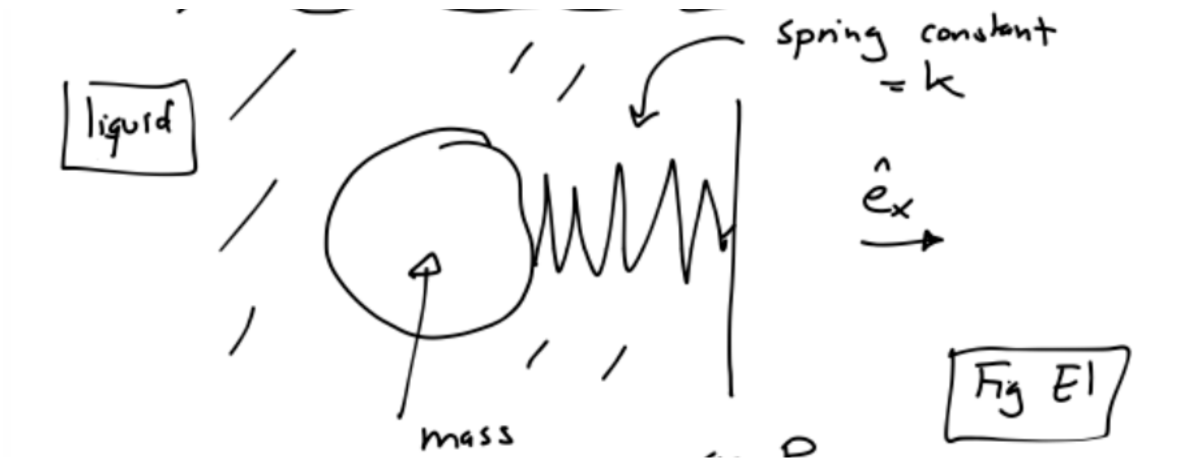


Figure 9: A bead attached to a spring in a viscous liquid.

The restoring force from the spring is

$$F_{\text{spring}} = -kx \tag{55}$$

where $k =$ spring constant and

$$x = \text{displacement from equilibrium.}$$

At low Reynolds numbers the viscous drag force is given by Stokes law:

$$F_{\text{drag}} = -\gamma v \tag{56}$$

where $\gamma = 6\pi\eta R$ for a spherical particle of radius R.

Newton's equation now becomes

$$m\frac{dv}{dt} = -kx - \gamma v \tag{57}$$

where

$$v = \frac{dx}{dt}.$$

At first sight this looks like the familiar damped harmonic oscillator. However, for microscopic systems in liquids,

$$\text{Re} \ll 1,$$

and inertial effects are typically negligible. In other words, the term mdv/dt is much smaller than the viscous drag term. We therefore set the particle inertia term

$$m\frac{dv}{dt} \approx 0.$$

A simple argument for this choice is that the mass m is proportional to the particle size volume, i.e., $m \propto L^3$, where the friction coefficient is $\gamma \propto L$. Thus, for sufficiently small L the Stokes law guarantee that the inertia term is negligible.

Equation (57) then gives

$$\gamma \frac{dx}{dt} = -kx \tag{58}$$

or equivalently

$$\frac{dx}{dt} = -\frac{k}{\gamma} x. \tag{59}$$

This equation has the solution

$$x(t) = x_0 e^{-t/\tau} \tag{60}$$

where

$$\tau = \frac{\gamma}{k} \tag{61}$$

is the relaxation time.

Thus, unlike ordinary macroscopic oscillators in air (solutions involving sinus and cosines), microscopic particles in liquids typically do not oscillate. Instead, they relax exponentially toward equilibrium.

This is an example of "Aristotle dynamics":

$$F \propto v$$

rather than

$$F = ma.$$

Back in the day, Aristotle, without being bother by empirical observations, proposed that "to have motion, we must have a force".

From a computational point of view, Eq. (59) is one of the simplest overdamped dynamical equations. A forward Euler update would be

```
x = x - (k/gamma)*x*dt
```

or, if thermal fluctuations are included,

```
x = x - (k/gamma)*x*dt + sqrt(2*D*dt)*np.random.randn()
```

where the diffusion constant D is connected to γ by the Einstein relation to be derived later in this section.

4.6 Low Reynolds number swimming and the scallop theorem

At low Reynolds numbers the fluid equations are linear and time independent:

$$0 = -\nabla p + \eta \nabla^2 \mathbf{V}.$$

There is no inertial term which remembers the past motion of the fluid. This has a surprising consequence: the equations are reversible in time. More precisely, if

$$\mathbf{V}(\mathbf{r}, t)$$

is a solution, then the reversed motion

$$-\mathbf{V}(\mathbf{r}, -t)$$

is also a solution. The fluid does not care whether we play the movie forward or backward, provided that we also reverse the velocities.

This has important consequences for swimming at low Reynolds numbers. At low Reynolds numbers, a swimmer performing reciprocal motion cannot obtain net propulsion. This statement is known as Purcell's scallop theorem.

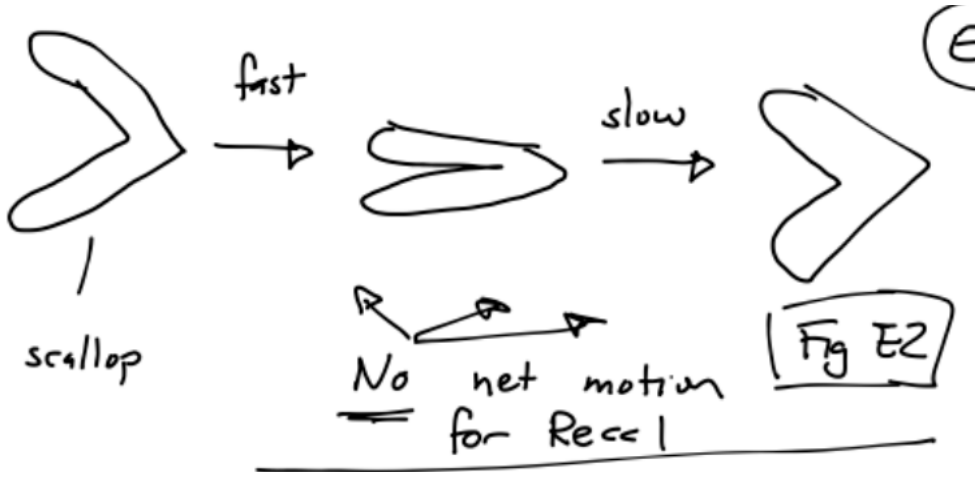


Figure 10: Reciprocal motion and Purcell's scallop theorem.

A scallop opens and closes symmetrically:

$$\text{open} \to \text{close}.$$

If this motion is then reversed,

$$\text{close} \to \text{open},$$

the fluid motion exactly reverses as well. The second half of the stroke undoes the first half. Hence there is no net displacement.

This is very different from swimming at high Reynolds number. A human swimmer can push water backward and then coast slightly because inertia is important. A bacterium cannot use such a strategy. At low Reynolds number, to swim you must perform a deformation cycle which is not identical to its own time reverse.

Microorganisms must therefore use non-reciprocal motion strategies. Examples include:

- rotating helical flagella,
- traveling waves along cilia,
- flexible body deformations.

The lesson is simple but deep: at low Reynolds number, motion requires geometry and timing. It is not enough to move back and forth; the motion must trace out a non-reversible cycle. You can read more about this in the excellent and easy.to-read article:

Purcell, “Life at low Reynolds numbers”, 1977.

4.7 The Einstein relation

In Section 3, we encountered diffusion. In the present section, we have so far studied the macroscopic motion of liquids and seen that a sphere moving through a viscous liquid has a friction coefficient $\gamma = 6\pi\eta R$. We now connect these two ideas. The result is the Einstein relation:

$$D = \frac{k_B T}{\gamma} \tag{62}$$

where γ is the friction coefficient. This equation is remarkable. The left-hand side describes random thermal motion. The right-hand side contains the temperature and the friction coefficient, which tells us how strongly the surrounding liquid resists motion. Thus the Einstein relation connects fluctuations and dissipation.

Let us derive it.

Consider a system at equilibrium in an external potential $U(x)$. Until now, we have mostly considered free diffusion, corresponding to $U(x) = 0$. Now we allow the particle to feel a force,

$$F(x) = -\frac{dU(x)}{dx}. \tag{63}$$

At low Reynolds number, force and velocity are related by

$$F = \gamma v.$$

Therefore, a force $F(x)$ gives a drift velocity

$$v(x) = \frac{F(x)}{\gamma}.$$

Here γ is again the friction coefficient. It can be measured experimentally by applying a known force and measuring the resulting velocity. The diffusion equation is now modified by a drift term:

$$\frac{\partial P(x,t)}{\partial t} = D\frac{d^2}{dx^2}P(x,t) - \frac{1}{\gamma}\frac{d}{dx}\left[FP(x,t)\right] \tag{64}$$

This is the diffusion equation with drift.

At steady-state and equilibrium,

$$\frac{\partial P(x,t)}{\partial t} = 0.$$

Hence, Eq. (64) gives

$$0 = D\frac{\partial^2 P(x)}{\partial x^2} - \frac{1}{\gamma}\frac{\partial}{\partial x}\left[F(x)P(x)\right]. \tag{65}$$

This equation has the solution

$$P(x) = \text{constant} \times e^{-U(x)/(\gamma D)}. \tag{66}$$

as is easy to check.

Now, connecting back to earlier sections, we know that at equilibrium, the probability density must also be given by the Boltzmann distribution,

$$P(x) = \text{constant} \times e^{-U(x)/(k_B T)}. \tag{67}$$

Comparing Eq. (66) and (67) we see that we must have

$$D = \frac{k_B T}{\gamma}. \tag{68}$$

This is the Einstein relation. The result also gives a useful physical intuition. Hotter systems diffuse faster than colder systems. And, combined with the Stokes law we have that larger particle diffuse slower and higher viscosity leads to a smaller diffusion constants.

Equipped with Stokes law and the Einstein relation, we can estimate the diffusion constant for a large globular protein. Take

$$R \sim 10 \text{ nm} = 10^{-8} \text{ m},$$

$$k_B = 1.38 \times 10^{-23} \ \frac{\text{m}^2\text{kg}}{\text{s}^2\text{K}},$$

$$T = 300 \text{ K} \sim 10^2 \text{ K},$$

and

$$\eta = 10^{-3} \text{ Pa s}.$$

Using $\gamma = 6\pi\eta R$, and ignoring numerical factors in an order-of-magnitude estimate,

$$D = \frac{k_B T}{\gamma} \sim \frac{k_B T}{\eta R}.$$

Thus,

$$D \sim \frac{10^{-23} \cdot 10^2}{10^{-3} \cdot 10^{-8}} \sim 10^{-10} \ \frac{\text{m}^2}{\text{s}}.$$

This is how we obtained the estimate used in the Diffusion section.

5 Molecular Motors

Cells contain a wide variety of molecular machines that convert chemical energy into mechanical work. These machines are not macroscopic engines made smaller. They operate in a world where $\mathrm{Re} \ll 1$ where inertia is negligible, where thermal fluctuations are large, and where motion is intrinsically stochastic.

One of the most important classes of such machines is the translational molecular motors. These motors move along filamentous tracks and are responsible for processes such as intracellular transport, muscle contraction, chromosome segregation, and cilia and flagella motion.

The central physical question in this section is:

How can directed motion arise in a noisy microscopic world?

At first sight this seems puzzling. In the diffusion section, we saw that thermal motion leads to random wandering. In the hydrodynamics section, we saw that microscopic motion through water is strongly overdamped. Yet inside cells, molecular motors walk directionally over long distances, pull cargo, generate forces, and organize cellular structure.

The key idea is that translational molecular motors are stochastic nonequilibrium machines. They do not eliminate Brownian motion. Instead, they use chemical energy, typically from ATP hydrolysis, to bias and rectify thermal fluctuations. Since the motion occurs along tracks, the dynamics of translational molecular motors is different than the motion of bacteria which makes use of non-reciprocal deformations for propulsion.

5.1 The speed of molecular motors

Motor proteins transport vesicles, organelles, proteins, and mRNA through cells. An important example occurs in neurons, where kinesin and dynein transport cargo over distances that can reach meters in large animals. Typical transport speeds are approximately

$$1 - 5\ \mu\mathrm{m/s}.$$

Motor proteins also play essential roles during cell division, where they help organize and move chromosomes using the mitotic spindle.

5.2 Motor families

The three main families of translational cytoskeletal motors are (i) myosin (ii) kinesin (iii) dynein. Myosin motors move on actin filaments, while kinesin and dynein move on microtubules.

Both actin filaments and microtubules are polar structures. This means that the two ends of the filament are physically and chemically different. Because the tracks have polarity, motor proteins can have a preferred direction of motion.

Typically (i) kinesin moves toward the plus end of microtubules (ii) dynein moves toward the minus end of microtubules (iii) most myosins move toward the plus end of actin filaments.

The biological details differ between motor families, but the physical themes are common: a molecular-scale object binds to a track, undergoes chemical transitions, changes conformation, and takes stochastic steps.

5.3 Motor architecture

A common structural feature of many molecular motors is the presence of a motor head domain. The motor head

- binds ATP,
- hydrolyzes ATP,
- binds to the filament track,
- undergoes conformational changes.

The conformational changes generated by ATP hydrolysis are amplified into mechanical motion. Many motors possess two motor heads. At the opposite end of the molecule, a tail domain binds cargo.

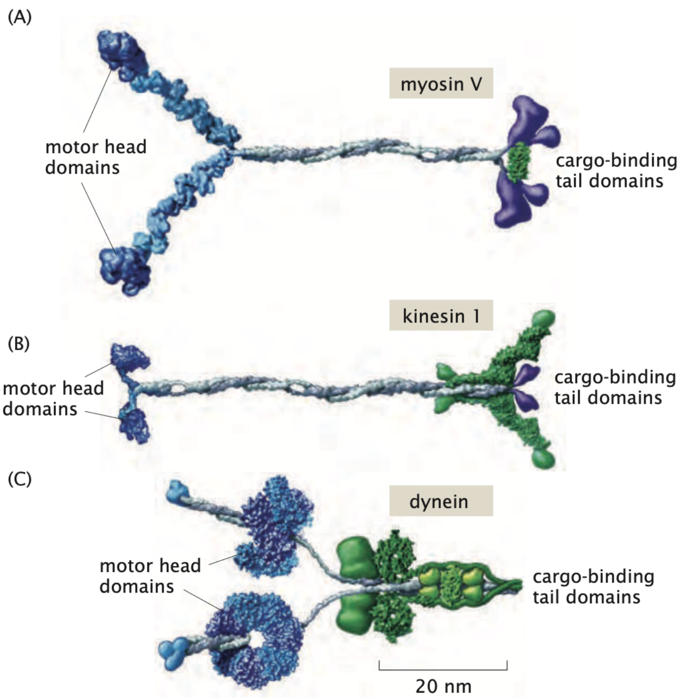


Figure 11: Translational motors: myosin, kinesin and dynein. The motor head domains binds to a track (actin or microtubules). From Phillips et al., *Physical Biology of the Cell*, 2nd ed.

This architecture is useful to keep in mind when we later introduce internal states. A motor is not just a point particle hopping on a line. It has chemical and conformational degrees of freedom. Nevertheless, a point-particle hopping model is often a very good zeroth-order physical description.

5.4 Discrete stepping

Motor proteins move by taking discrete steps along their filament tracks. Each step is coupled to a biochemical cycle involving ATP binding, ATP hydrolysis, phosphate release, and ADP release.

The motor velocity can be written schematically as

$$v = (\text{step size}) \times (\text{steps per second}). \tag{69}$$

This is a simple equation, but it contains much of the physics. A motor can move faster either by taking larger steps or by stepping more often. For kinesin, the step size is of order several nanometers, so cellular transport over micrometer or millimeter distances requires many repeated stochastic stepping events.

An important property of motors is their processivity, namely the number of consecutive steps a motor can take before detaching from the filament. A highly processive motor can walk for many steps without falling off its track. This is essential for long-distance transport.

5.5 Myosin walking and hand-over-hand motion

One of the best studied examples is myosin motion on actin filaments. The myosin motor is responsible for muscle contraction and cargo transport inside cells. A simplified biochemical cycle is:

1. myosin attached to actin,
2. ATP binding causes release,
3. ATP hydrolysis "cocks" the head,
4. phosphate release permits rebinding,
5. ADP release generates the power stroke.

Experiments using single-molecule fluorescence and AFM imaging have demonstrated that motors such as myosin V move by alternating the positions of their two motor heads. This mechanism is often referred to as "hand-over-hand" motion.

5.6 Motors as biased random walkers

We now turn from biological description to physical modeling.

At mesoscopic scales, thermal fluctuations are unavoidable. Molecular motors therefore operate in a noisy environment, and their motion should be viewed as stochastic rather than deterministic.

A useful starting point is to model a molecular motor as a random walker on a one-dimensional lattice with lattice spacing a. The lattice sites represent binding sites along the filament. The dynamics is the same as in the DIffusion section, but with one important difference: in general,

$$k_+ \neq k_-$$

The motor is a biased random walker. It still fluctuates, but the fluctuations have a preferred direction. How can we get have different forward and backward jump rates? ATP hydrolysis provides the necessary free-energy input. Very roughly, the chemical reaction

$$\mathrm{ATP} \to \mathrm{ADP} + \mathrm{P_i}$$

releases free energy, and the motor uses this free energy to bias its mechanochemical cycle.

The dynamics is governed by Eq. (24). If we write $p(x, t+\Delta t) - p(x, t) \approx \Delta t p(x, t)$ and take $\Delta t \to 0$, we get that the motor dynamics is are governed by the master equation:

$$\frac{dP_n}{dt} = k_+ P_{n-1} + k_- P_{n+1} - (k_+ + k_-)P_n. \tag{70}$$

This equation described how Probability "flows2 into site n from neighboring sites, and probability flows out of site n through forward and backward jumps. The first term,

$$k_+ P_{n-1},$$

corresponds to motors arriving at site n by stepping forward from site $n-1$. The second term,

$$k_- P_{n+1},$$

corresponds to motors arriving at site n by stepping backward from site $n+1$. The final term,

$$-(k_+ + k_-)P_n,$$

is the loss of probability from site n.

This is the simplest stochastic model of directed motor motion.

5.7 Mean displacement and velocity

The mean position of the motor is

$$\langle x(t) \rangle = a \sum_n n P_n(t). \tag{71}$$

Differentiating with respect to time gives

$$\frac{d\langle x \rangle}{dt} = a \sum_n n \frac{dP_n}{dt}.$$

Using the master equation,

$$\frac{d\langle x \rangle}{dt} = a \sum_n n \left[k_+ P_{n-1} + k_- P_{n+1} - (k_+ + k_-)P_n \right].$$

We now shift summation indices in the first two terms. In the first term let $m = n - 1$, so that $n = m + 1$. Then

$$\sum_n nP_{n-1} = \sum_m (m+1)P_m = \sum_m mP_m + \sum_m P_m.$$

Since probabilities are normalized,

$$\sum_m P_m = 1.$$

Similarly, in the second term let $m = n + 1$, so that $n = m - 1$. Then

$$\sum_n nP_{n+1} = \sum_m (m-1)P_m = \sum_m mP_m - \sum_m P_m.$$

Putting this into the expression for the mean velocity gives

$$\frac{d\langle x\rangle}{dt} = a\left[k_+\left(\sum_m mP_m + 1\right) + k_-\left(\sum_m mP_m - 1\right) - (k_+ + k_-)\sum_m mP_m\right].$$

The terms involving $\sum_m mP_m$ cancel, and we obtain

$$\frac{d\langle x\rangle}{dt} = a(k_+ - k_-). \tag{72}$$

Hence the average motor velocity is

$$v = a(k_+ - k_-). \tag{73}$$

Directed motion therefore emerges from a bias in the transition rates. This is a very compact physical result: the motor moves forward on average if forward steps occur more often than backward steps.

5.8 Diffusion coefficient of a motor

Even in the presence of drift, fluctuations remain important. Two motors with the same average velocity will not follow identical trajectories.

The variance of the position is

$$\langle x^2\rangle - \langle x\rangle^2. \tag{74}$$

Proceeding similarly to the calculation of the mean displacement yields the effective diffusion coefficient

$$D = \frac{a^2}{2}(k_+ + k_-). \tag{75}$$

Thus the motor simultaneously exhibits deterministic drift and stochastic diffusion. The drift is controlled by the difference of the rates, $k_+ - k_-$, whereas the fluctuations are controlled by the sum of the rates, $k_+ + k_-$.

5.9 Continuum limit

To obtain a continuum description, introduce a continuous probability density $\rho(x,t)$, where

$$x = na.$$

Proceeding as in the section Diffusion, expanding the master equation to second order in a gives the drift–diffusion equation

$$\frac{\partial \rho}{\partial t} = -v\frac{\partial \rho}{\partial x} + D\frac{\partial^2 \rho}{\partial x^2}. \tag{76}$$

This is a Fokker–Planck equation describing biased diffusion.

The first term on the right-hand side describes directed motion with velocity v. The second term describes diffusive spreading around the mean position.

For an initially localized motor,

$$\rho(x,0) = \delta(x - x_0), \tag{77}$$

the solution is

$$\rho(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left[-\frac{(x - x_0 - vt)^2}{4Dt}\right]. \tag{78}$$

The distribution therefore drifts with velocity v and broadens diffusively with width

$$\sim \sqrt{Dt}.$$

This is exactly what we expect from a stochastic motor. On average, it moves forward. But around that average motion, there are fluctuations.

5.10 Force–velocity relations

Molecular motors slow down when subjected to external loads. This is the microscopic analogue of pushing a machine against a resisting force. The larger the load, the harder it is for the motor to step forward.

Different motors exhibit different force–velocity curves.

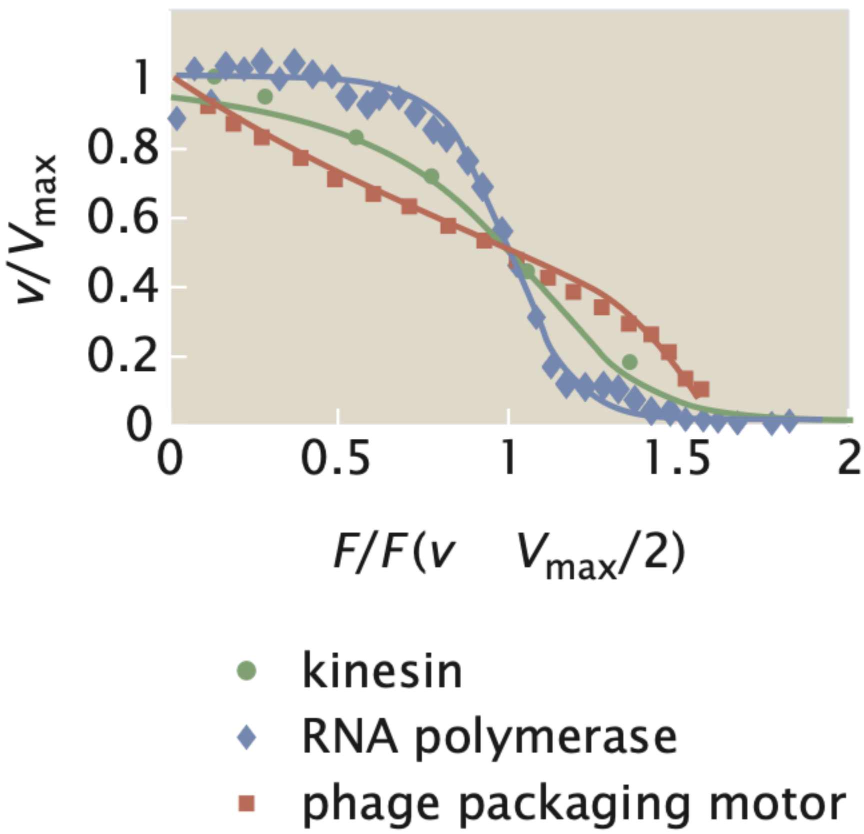


Figure 12: Force–velocity figure determined through experiments. From Phillips et al., *Physical Biology of the Cell*, 2nd ed.

These force–velocity relations reflect the detailed mechanism by which chemical energy is converted into mechanical work.

In the simple biased-random-walk model, the effect of an external force can be included by making the transition rates force dependent. Suppose the motor works against an external force F. A common phenomenological form is to assume that the force enters through a Boltzmann factor:

$$k_+(F) = k_+^{(0)} e^{-\theta F a / k_B T}, \tag{79}$$

and

$$k_-(F) = k_-^{(0)} e^{(1-\theta) F a / k_B T}, \tag{80}$$

where

$$0 \le \theta \le 1$$

describes how the load affects the transition state.

The motor velocity then becomes

$$v(F) = a \left[k_+(F) - k_-(F) \right]. \tag{81}$$

As the load increases, the forward rate decreases and the backward rate increases. Therefore the velocity decreases.

The stall force (the force at which the motor velocity becomes zero) is obtained from

$$v(F_{\text{stall}}) = 0. \tag{82}$$

At stall, the motor still undergoes microscopic stochastic transitions, but there is no net average motion.

5.11 Numerical simulation of motor trajectories

The biased random-walk model is also a useful playground for simulations. A minimal stochastic simulation procedure is:

```
n = 0
t = 0

while t < T:
    R = np.random.rand()

    if R < k_plus*dt:
        n = n + 1
    elif R < (k_plus + k_minus)*dt:
        n = n - 1

    t = t + dt
```

If we instead consider time as continuous, one must make use of the Gillespie algorithm. In that approach, the waiting time to the next event is drawn from an exponential distribution with rate

$$k_{\text{tot}} = k_+ + k_-.$$

The waiting time is

$$\tau = -\frac{1}{k_{\text{tot}}} \ln R_1,$$

where R_1 is a uniform random number between 0 and 1. A second random number R_2 is used to decide whether the event is a forward or backward step:

$$\text{forward step} \quad \text{if} \quad R_2 < \frac{k_+}{k_+ + k_-}.$$

The corresponding pseudocode is:

```
n = 0
t = 0

while t < T:
```

```
ktot = k_plus + k_minus
tau = -np.log(np.random.rand())/ktot
t = t + tau

if np.random.rand() < k_plus/ktot:
    n = n + 1
else:
    n = n - 1
```

This kind of simulation is a direct numerical realization of the master equation.

5.12 The two-state model*

The remaining subsections provide somewhat more detailed mathematical models. They are useful for connecting molecular motors to stochastic processes and nonequilibrium statistical mechanics, but they can be skipped in a first reading.

More realistic descriptions include internal biochemical states. In the two-state model, the motor alternates between two internal states:

- state 1,
- state 2.

These states may represent different conformations of the motor during the ATP hydrolysis cycle, for example a strongly bound state and a weakly bound state.

Let

$$P_n^{(1)}(t) \tag{83}$$

and

$$P_n^{(2)}(t) \tag{84}$$

denote the probabilities for occupying site n in the two states.

The motor can step forward and backward within each internal state. We denote the forward and backward rates in state 1 by

$$u_1, \quad w_1,$$

and the forward and backward rates in state 2 by

$$u_2, \quad w_2.$$

The motor can also switch between the internal states, with transition rates

$$k_{12}$$

from state 1 to state 2, and

$$k_{21}$$

from state 2 to state 1.

The coupled master equations become

$$\frac{dP_n^{(1)}}{dt} = u_1 P_{n-1}^{(1)} + w_1 P_{n+1}^{(1)} + k_{21} P_n^{(2)} \\ - (u_1 + w_1 + k_{12}) P_n^{(1)}, \tag{85}$$

and

$$\frac{dP_n^{(2)}}{dt} = u_2 P_{n-1}^{(2)} + w_2 P_{n+1}^{(2)} + k_{12} P_n^{(1)} \\ - (u_2 + w_2 + k_{21}) P_n^{(2)}. \tag{86}$$

These equations look more complicated than the one-state master equation, but the logic is the same. Probability flows into and out of each site and each internal state.

The new feature is that probability can also move between internal states. This is how chemistry enters the mechanical stepping model.

The two-state model is useful because it makes explicit that a motor is not just moving in space. It is moving in a combined space of position and chemical state.

6 Rate equations and dynamics in the cell

In the previous sections we studied the motion of *single molecules* at their diffusion and hydrodynamics. We now turn to the motion of many molecules. Instead of following every individual molecule, we will describe the system using concentrations and rate equations.

We will study the important examples:

- irreversible molecular transformations,
- reversible transformations,
- bimolecular reactions,
- enzyme kinetics and the Michaelis–Menten equation.

Many of these models are among the simplest examples of dynamical systems in biology. They are also excellent examples of how physical reasoning, probability, and chemistry fit together.

6.1 The rate equation paradigm

In the diffusion section, we studied changes in concentration caused by molecular motion. Here we instead focus on transformations between molecular species. Examples include molecular decay, conformational switching, ion-channel opening and closing, binding and unbinding reactions, and enzymatic catalysis.

The central assumption of the rate-equation approach is the *perfect-mixing assumption.* We assume that the system is well-stirred, so that concentrations depend only on time and not on position:

$$C_i(\mathbf{r}, t) \to C_i(t).$$

Thus, spatial inhomogeneities are neglected.

This approximation is often good when diffusion is sufficiently fast. Very roughly, diffusion must mix the system faster than reactions consume reactants. However, the approximation is not always valid. In low dimensions, spatial fluctuations can become important. Reaction–diffusion systems can then behave very differently from perfectly mixed systems. From now on we ignore this subtlety and assume the system at hand is always well-mixed.

A general reaction network is described by concentrations

$$C_1(t), C_2(t), \ldots$$

satisfying coupled ordinary differential equations of the form

$$\frac{dC_i}{dt} = F_i(C_1, C_2, \ldots; k_1, k_2, \ldots). \tag{87}$$

Here the parameters

$$k_1, k_2, \ldots$$

are rate constants. The challenge we will tackle is to write down explicit form for $F_i(\ldots)$ for a few prototype processes.

6.2 Irreversible decay*

Consider the irreversible transformation

$$A \to B. \tag{88}$$

We imagine that molecules of type A transform into molecules of type B with rate constant k. During a short time interval Δt, a fraction

$$k\Delta t$$

of the A molecules transform into B molecules. The concentration therefore changes according to

$$C_A(t + \Delta t) - C_A(t) = -k\Delta t\, C_A(t).$$

Dividing by Δt and taking the limit $\Delta t \to 0$ gives

$$\frac{dC_A}{dt} = -kC_A. \tag{89}$$

This is one of the simplest differential equations in physics. In this case we have that $F(C_A, k) = -kC_A$.

The solution of the equation above is

$$C_A(t) = C_{A,0}e^{-kt}. \tag{90}$$

The concentration therefore decays exponentially. The characteristic time is $\tau = 1/k$. Large k means rapid decay, while small k means slow decay.

This exponential form appears everywhere in physics and biology. It is a signature of a process with a constant probability per unit time of occurring.

The rate equation above can be understood microscopically. Suppose there are initially $N_A(0)$ molecules of type A in a volume V. Define the probability that a particular molecule decays during a short interval Δt as

$$q = k\Delta t. \tag{91}$$

If the molecules behave independently, then after one time step the expected number of surviving molecules is

$$N_A(\Delta t) = N_A(0)(1 - k\Delta t).$$

After n time steps,

$$N_A(t) = N_A(0)(1 - k\Delta t)^n$$

with

$$t = n\Delta t.$$

Using

$$e^{-x} = \lim_{n\to\infty} \left(1 - \frac{x}{n}\right)^n,$$

we obtain

$$N_A(t) = N_A(0)e^{-kt}.$$

Dividing by the volume gives Eq. (90).

Thus the deterministic rate equation emerges from many independent random microscopic events.

6.3 Reversible transformations

Many biological processes are reversible. Consider therefore

$$A \underset{k_-}{\overset{k_+}{\rightleftarrows}} B. \tag{92}$$

The concentration of A changes because A molecules transform into B with rate k_+, while B molecules transform into A with rate k_-.

The rate equation is

$$\frac{dC_A}{dt} = -k_+ C_A + k_- C_B. \tag{93}$$

Particle conservation implies

$$C_A(t) + C_B(t) = C_0. \tag{94}$$

Substituting

$$C_B = C_0 - C_A$$

into Eq. (93) gives

$$\frac{dC_A}{dt} = -(k_+ + k_-)C_A + k_- C_0.$$

The linear equation above has solution

$$C_A(t) = \frac{k_-}{k_+ + k_-}C_0 + Fe^{-(k_+ + k_-)t}. \tag{95}$$

as is easy to check.

If initially all molecules are of type A,

$$C_A(0) = C_0,$$

then

$$C_A(t) = \frac{k_-}{k_+ + k_-}C_0 + \frac{k_+}{k_+ + k_-}C_0 e^{-(k_+ + k_-)t}. \tag{96}$$

At long times,

$$C_A(\infty) = \frac{k_-}{k_+ + k_-}C_0. \tag{97}$$

The approach to equilibrium is exponential, with relaxation rate

$$k_+ + k_-.$$

Notice something important. The equilibrium concentrations depend on the ratio

$$\frac{k_+}{k_-},$$

whereas the relaxation speed depends on the sum

$$k_+ + k_-.$$

This distinction appears repeatedly in stochastic physics.

At equilibrium, the ratio of concentrations is determined by the free energy difference between the states.

Suppose state B differs from state A by a free-energy difference

$$\Delta G.$$

Equilibrium statistical mechanics gives

$$\frac{C_B}{C_A} = e^{-\Delta G/k_B T}. \tag{98}$$

Comparing with the equilibrium solution of the rate equations gives

$$\frac{k_+}{k_-} = e^{-\Delta G/k_B T}. \tag{99}$$

Thus the ratio of forward and backward rates is constrained by the free energy landscape.

6.4 Bimolecular reactions*

We now move beyond single-particle transformations and consider reactions requiring encounters between molecules.

The simplest example is

$$A + B \to AB. \tag{100}$$

where here AB denotes a complex formed by A and B.

The reversible case is

$$A + B \underset{k_{\mathrm{off}}}{\overset{k_{AB}}{\rightleftarrows}} AB. \tag{101}$$

This kind of process is everywhere in biology. Examples include receptor–ligand binding, DNA hybridization, protein complex formation, transcription-factor binding, and enzyme–substrate binding.

The key physical difference compared with single-molecule decay is that two molecules must first find each other.

To argue for the rate equation for the process of complex formation (and dissociation), we divide the system into many small volume elements, or voxels. A reaction can occur only if an A molecule and a B molecule occupy the same local region.

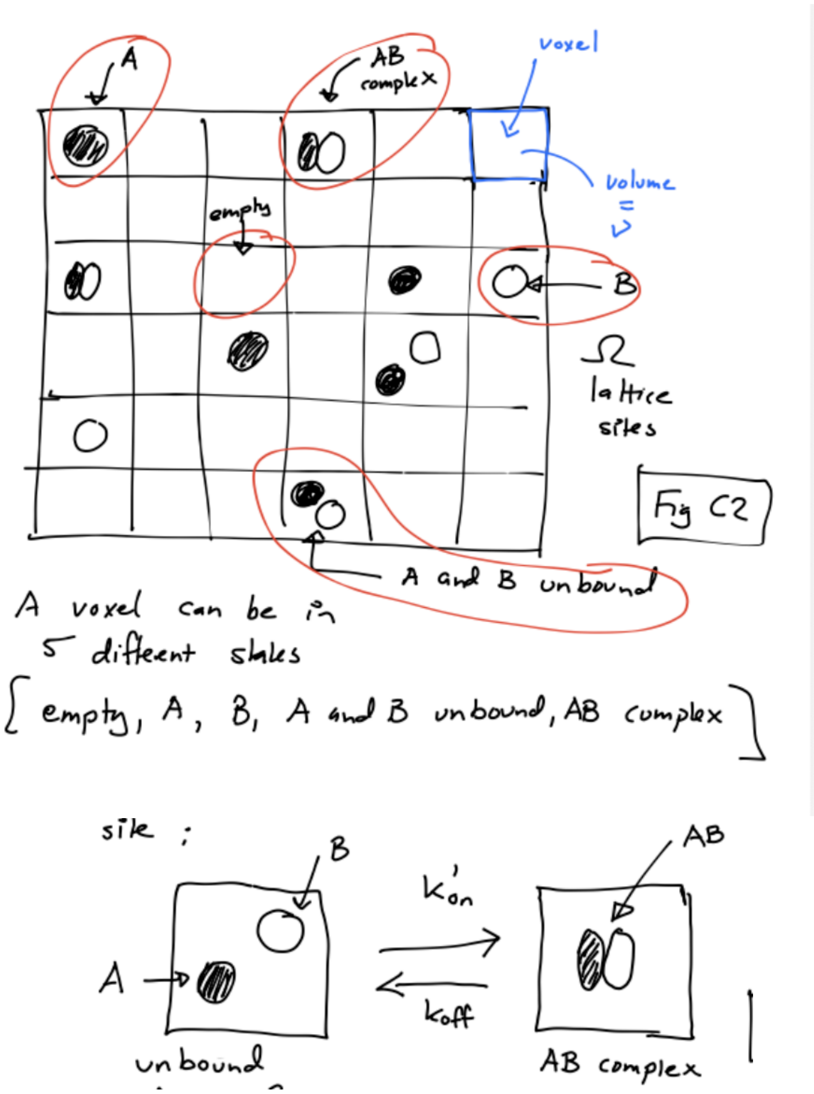


Figure 13: Two freely diffusing molecules, A and B, can associate to form a complex AB, while the complex can also dissociate back into free molecules. (Top) An AB complex can form only when an A and a B molecule are in the same voxel. (Bottom) Binding and unbinding happens within a voxel.

Each voxel can be in one of several states: empty, containing only A, containing only B, containing an unbound A and B pair, or containing an AB complex.

Now let

$$C_A = [A], \qquad C_B = [B], \qquad C_{AB} = [AB].$$

where $[X]$ denotes "the concentration of molecules of type X". The concentration of complexes changes because complexes dissociate with rate k_{off}, while free molecules bind with rate k_{AB}.

The rate equation is therefore

$$\frac{dC_{AB}}{dt} = -k_{\text{off}} C_{AB} + k_{AB} C_A C_B. \tag{102}$$

Notice the nonlinear term

$$C_A C_B.$$

This appears because the reaction requires an encounter between two molecules (two molecules in the same voxel). The (equilibrium) the probability of finding an A and a B in the same voxel is proportional to the number of molecules of type A multiplied by the number of type B molecules. NOTE: This statement is only true of we assume that the perfect mixing.

The concentrations of free molecules satisfy

$$\frac{dC_A}{dt} = +k_{\text{off}} C_{AB} - k_{AB} C_A C_B, \tag{103}$$

$$\frac{dC_B}{dt} = +k_{\text{off}} C_{AB} - k_{AB} C_A C_B. \tag{104}$$

These equations can be solved numerically, and in special cases also analytically. For the lectures, the most important point is the physical origin of the nonlinear term and the conservation laws.

The units of the rate constants are worth examining carefully. Since

$$\frac{dC_{AB}}{dt} \sim k_{AB} C_A C_B,$$

we see that

$$[k_{AB}] = \frac{1}{\text{concentration} \times \text{time}}. \tag{105}$$

In contrast,

$$[k_{\text{off}}] = \frac{1}{\text{time}}. \tag{106}$$

This difference reflects the underlying physics. Dissociation is a single-complex process. Association requires two molecules to encounter one another.

At equilibrium,

$$\frac{dC_{AB}}{dt} = 0.$$

Equation (102) then gives

$$k_{\text{off}} C_{AB}^{\text{eq}} = k_{AB} C_A^{\text{eq}} C_B^{\text{eq}}. \tag{107}$$

Hence,

$$\frac{C_A^{\text{eq}} C_B^{\text{eq}}}{C_{AB}^{\text{eq}}} = \frac{k_{\text{off}}}{k_{AB}} \equiv K_d. \tag{108}$$

This is the law of mass action.

6.5 Numerical solution of nonlinear rate equations*

Unlike the simple decay equations, nonlinear bimolecular equations are not always easy to solve analytically. However, they are straightforward to solve numerically using one of the methods from Appendix C. When solving systems of of equations of the form above, one should always check convergence with respect to the time step Δt, i.e., make Δt smaller and smaller until the concentrations are unaltered.

6.6 Enzyme kinetics*

Enzymes are catalytic molecules that accelerate chemical reactions.

An enzyme binds a substrate, forms an intermediate complex, and then releases product. The simplest model is

$$E + S \underset{k_-}{\overset{k_+}{\rightleftarrows}} ES \xrightarrow{r} E + P. \tag{109}$$

Here E is the enzyme, S is the substrate, ES is the enzyme–substrate complex, and P is the product.

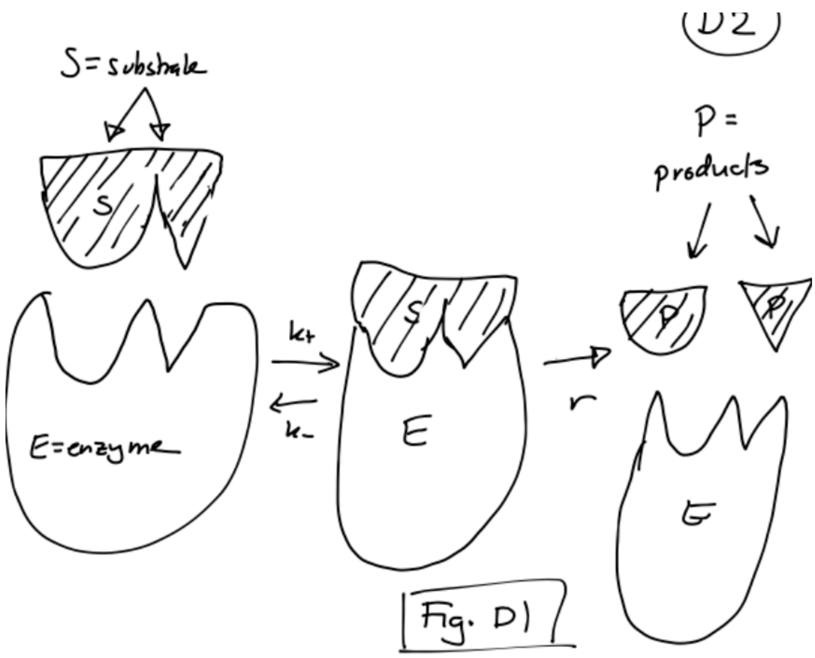


Figure 14: Enzyme reaction $E + S \rightleftarrows ES \to E + P$. The enzyme E binds substrate S to form the enzyme–substrate complex ES, which can either dissociate or produce product P while releasing the enzyme.

The corresponding rate equations are

$$\frac{d[E]}{dt} = -k_+[E][S] + k_-[ES] + r[ES], \tag{110}$$

$$\frac{d[S]}{dt} = -k_+[E][S] + k_-[ES], \tag{111}$$

$$\frac{d[ES]}{dt} = +k_+[E][S] - k_-[ES] - r[ES], \tag{112}$$

$$\frac{d[P]}{dt} = r[ES]. \tag{113}$$

This is the full deterministic model for the simple enzyme reaction. Eqs. (110)–(113) can be solved directly by the Euler method from Appendix C.

A common simplification of the equations above is the steady-state approximation. We assume that the intermediate complex concentration changes slowly:

$$\frac{d[ES]}{dt} \approx 0. \tag{114}$$

Equation (112) then gives

$$k_+[E][S] = (k_- + r)[ES]. \tag{115}$$

Define the Michaelis constant

$$K_m = \frac{k_- + r}{k_+}. \tag{116}$$

The total enzyme concentration is conserved:

$$[E]_{\text{tot}} = [E] + [ES]. \tag{117}$$

Thus,

$$[E] = [E]_{\text{tot}} - [ES].$$

Inserting this into Eq. (115) gives

$$k_+ \left([E]_{\text{tot}} - [ES]\right)[S] = (k_- + r)[ES].$$

Rearranging,

$$k_+[E]_{\text{tot}}[S] = (k_- + r + k_+[S])[ES].$$

Using the definition of K_m, we obtain

$$[ES] = \frac{[E]_{\text{tot}}[S]}{K_m + [S]}. \tag{118}$$

The product formation rate is therefore

$$\frac{d[P]}{dt} = r[ES] = \frac{V_{\max}[S]}{K_m + [S]}, \tag{119}$$

where

$$V_{\max} = r[E]_{\text{tot}}. \tag{120}$$

Equation (119) is the Michaelis–Menten equation.

The Michaelis–Menten equation interpolates between two regimes:

For small substrate concentration,

$$[S] \ll K_m,$$

we obtain

$$\frac{d[P]}{dt} \approx \frac{V_{\max}}{K_m}[S].$$

The reaction rate is proportional to substrate concentration.

For large substrate concentration,

$$[S] \gg K_m,$$

we obtain

$$\frac{d[P]}{dt} \approx V_{\max}.$$

The enzyme becomes saturated. Increasing substrate concentration further cannot significantly increase the reaction speed.

The parameter K_m has a useful interpretation. When

$$[S] = K_m,$$

the reaction rate is half maximal:

$$\frac{d[P]}{dt} = \frac{V_{\max}}{2}.$$

Thus K_m is the substrate concentration at which the enzyme operates at half of its maximum speed.

A Generating random numbers

A.1 Random integers

Suppose we are interested in generating random numbers J which takes integer values $i = 1, 2, 3, ..., N$. Then simply:

$$J = 1 + [N \cdot R] \tag{121}$$

where $[x]$ is obtained by truncation of x (for instance, $[7.9] = 7$).

A.2 The Accept/Reject Method

The continuous transformation method is simple and convenient provided that a closed expression for C^{-1} can be obtained, but this is far from always the case. A more general approach, which does not require knowledge of C^{-1}, is the *accept/reject method*.

To generate random numbers with a given distribution $p(x)$, the accept/reject method method makes use of an auxiliary function $f_0(x)$, which must satisfy $f_0(x) \geq p(x)$. $f_0(x)$ should be chosen so that it is easy to obtain random numbers from the distribution:

$$p_0(x) = \frac{f_0(x)}{\int_{-\infty}^{\infty} f_0(x)dx} \tag{122}$$

[note that $f_0(x)$ itself is not a normalized distribution]. For a given $f_0(x)$, the method is as follows:

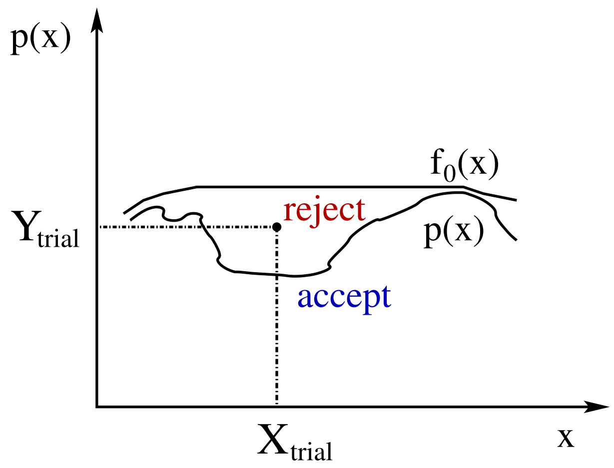
1. Draw a (trial) random number X_{trial} from the distribution $p_0(x)$.

2. Draw a uniform random number R between 0 and 1. Accept X_{trial} if

$$R < \frac{p(X_{\text{trial}})}{f_0(X_{\text{trial}})},$$

and reject otherwise.

In the scheme above we choose a random point $(X_{\text{trial}}, Y_{\text{trial}})$ [where $Y_{\text{trial}} = f_0(X_{\text{trial}})R$] uniformly in the area under $f_0(x)$, see figure below.



We then accept only those points (X, Y) which are in the area under $p(x)$. The X component of the accepted points has the distribution $p(x)$.

Example: Kahn's method for normally distributed random numbers.

Consider the distribution

$$p(x) = \begin{cases} 2\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

(if a random number with this distribution is given a random sign, a normally distributed random number is obtained). Use the accept/reject method with

$$f_0(x) = \begin{cases} \sqrt{\frac{2}{\pi}}\mathrm{e}^{-x+1/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

This choice of f_0 is OK since

$$\frac{p(x)}{f_0(x)} = \mathrm{e}^{-(x-1)^2/2} \leq 1\,,$$

and the corresponding probability density function is

$$p_0(x) = \frac{f_0(x)}{\int_{-\infty}^{\infty} f_0(x)dx} = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

We now follow the steps above:

1. Choose $X_{\text{trial}} = -\ln R_1$ where R_1 is a uniform random number $\in [0, 1]$ (exponential distribution; see previous example).

2. Accept X_{trial} if
$$R_2 < \frac{p(X_{\text{trial}})}{f_0(X_{\text{trial}})} = \mathrm{e}^{-(X_{\text{trial}}-1)^2/2}$$

How efficient is this method? This depends crucially on the acceptance rate, which is the area under $p(x)$ (which is 1) divided by the area under $f_0(x)$. In this particular example, we find an acceptance probability of

$$\frac{1}{\sqrt{\frac{2e}{\pi}}\int_0^{\infty}\mathrm{e}^{-x}dx} \approx 0.76\,.$$

A.3 The Central Limit Theorem - sums of independent and identically distributed random variables

Suppose $X_1, \ldots, X_N$ are *independent* and *identically distributed* [i.e. X_i are determined by the same $p(x)$] random variables (*iid* random variables, in short), with mean μ and variance σ^2. What is then the distribution of their sum? This question has a surprisingly simple answer in the limit $N \to \infty$.

This result, called the *central limit theorem* (CLT), says that the random variable

$$\tilde{S}_N = \frac{X_1 + ... + X_N - N\mu}{\sigma\sqrt{N}}$$

becomes normally distributed with zero mean and unit variance as $N \to \infty$ (provided that μ and σ^2 are finite).

In the derivation of the CLT one never looked at the precise form of the probability density function of the individual X_i's — the result holds irrespective of the precise form of this distribution. This makes the central limit theorem extremely useful.

Example: Consider the random walk problem from first section. The position of the random walker after n steps is a random variable $X = R_1 + R_2 + ... + R_n$, where R_i are random numbers. The central limit theorem tells us that the distribution $p_n(x)$ is

$$p_n(x) = \frac{1}{\sqrt{2\pi n\sigma^2}} \mathrm{e}^{-\frac{(x-n\mu)^2}{2n\sigma^2}} \tag{123}$$

We now need μ and σ^2. In the simulations the individual jumps were take to be of length a and equally probable to the left and right, i.e. we used a distribution $\tilde{p}(r) = [\delta(r-a) + \delta(r+a)]/2$. Calculating the first and second moment yields

$$\begin{aligned}\mu &= \int_{-\infty}^{\infty} r\tilde{p}(r)dx = 0 \\ \sigma^2 &= \int_{-\infty}^{\infty} r^2\tilde{p}(r)dx = a^2\end{aligned}$$

If we further introduce the time $t = n\Delta t$, and a diffusion constant $D \equiv a^2/(2\Delta t)$, Eq. (123) becomes:

$$p_n(x) = p(x,t|x_0 = 0) = \frac{1}{\sqrt{4\pi Dt}} \mathrm{e}^{-\frac{x^2}{4Dt}} \tag{124}$$

This result we recognize from previous Lectures: it is the solution to the one-dimensional diffusion equation:

$$\frac{\partial p(x,t|x_0)}{\partial t} = D\frac{\partial^2 p(x,t|x_0)}{\partial x^2}, \tag{125}$$

with initial condition $p(x,t|x_0) = \delta(x - x_0)$.

B Derivation of the Navier-Stokes equations for an incompressible fluid

B.1 Viscosity

Consider a liquid confined between two plates separated by a distance b. Assume that the lower plate is stationary and the upper plate moves with velocity $v_z(b)$.

For sufficiently small b the velocity profile becomes approximately linear:

$$\frac{v_z(b) - v_z(0)}{b} \approx \frac{\partial v_z}{\partial x}.$$

Newton proposed the constitutive relation

$$\frac{F_z}{A} = \eta \frac{\partial v_z}{\partial x}, \tag{126}$$

where:

- η is the viscosity,
- F_z/A is the shear stress,
- and $\partial v_z/\partial x$ is the velocity gradient.

A liquid satisfying Eq. (126) is called a Newtonian fluid. The SI unit of viscosity is

$$[\eta] = \mathrm{Pa\,s}.$$

For water:

$$\eta_{\text{water}} \approx 10^{-3}\ \mathrm{Pa\,s}.$$

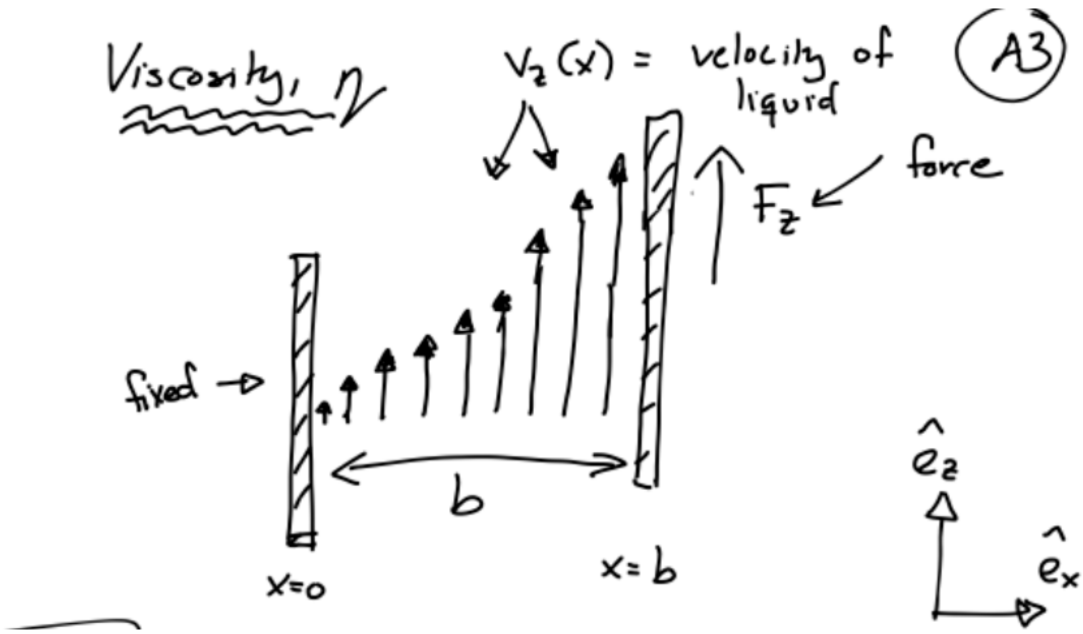


Figure 15: Moving-plate geometry illustrating viscosity and velocity gradients.

B.2 Incompressible fluids

We now derive the incompressibility condition.

Consider a small volume element with volume:

$$dV = bcd.$$

Assume that the side lengths depend on time.

We first investigate how the side length $d(t)$ changes during a short time interval Δt. At time $t + \Delta t$ we have

$$d(t + \Delta t) = d(t) + [v_z(z + d, t) - v_z(z, t)]\,\Delta t.$$

Using a Taylor expansion:

$$v_z(z+d,t) = v_z(z,t) + \frac{\partial v_z}{\partial z} d + \mathcal{O}(d^2),$$

which gives

$$d(t+\Delta t) = d(t)\left[1 + \frac{\partial v_z}{\partial z}\Delta t\right].$$

Applying the same reasoning to all directions gives

$$b(t+\Delta t) = b(t)\left[1 + \frac{\partial v_x}{\partial x}\Delta t\right],$$

$$c(t+\Delta t) = c(t)\left[1 + \frac{\partial v_y}{\partial y}\Delta t\right].$$

The volume element at time $t + \Delta t$ becomes

$$dV(t+\Delta t) = b(t+\Delta t)c(t+\Delta t)d(t+\Delta t).$$

Keeping only terms linear in Δt yields

$$dV(t+\Delta t) = dV(t)\left[1 + \Delta t\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)\right].$$

For an incompressible fluid the volume must remain constant:

$$dV(t+\Delta t) = dV(t).$$

Hence:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0. \tag{127}$$

Using vector notation:

$$\nabla \cdot \mathbf{v} = 0. \tag{128}$$

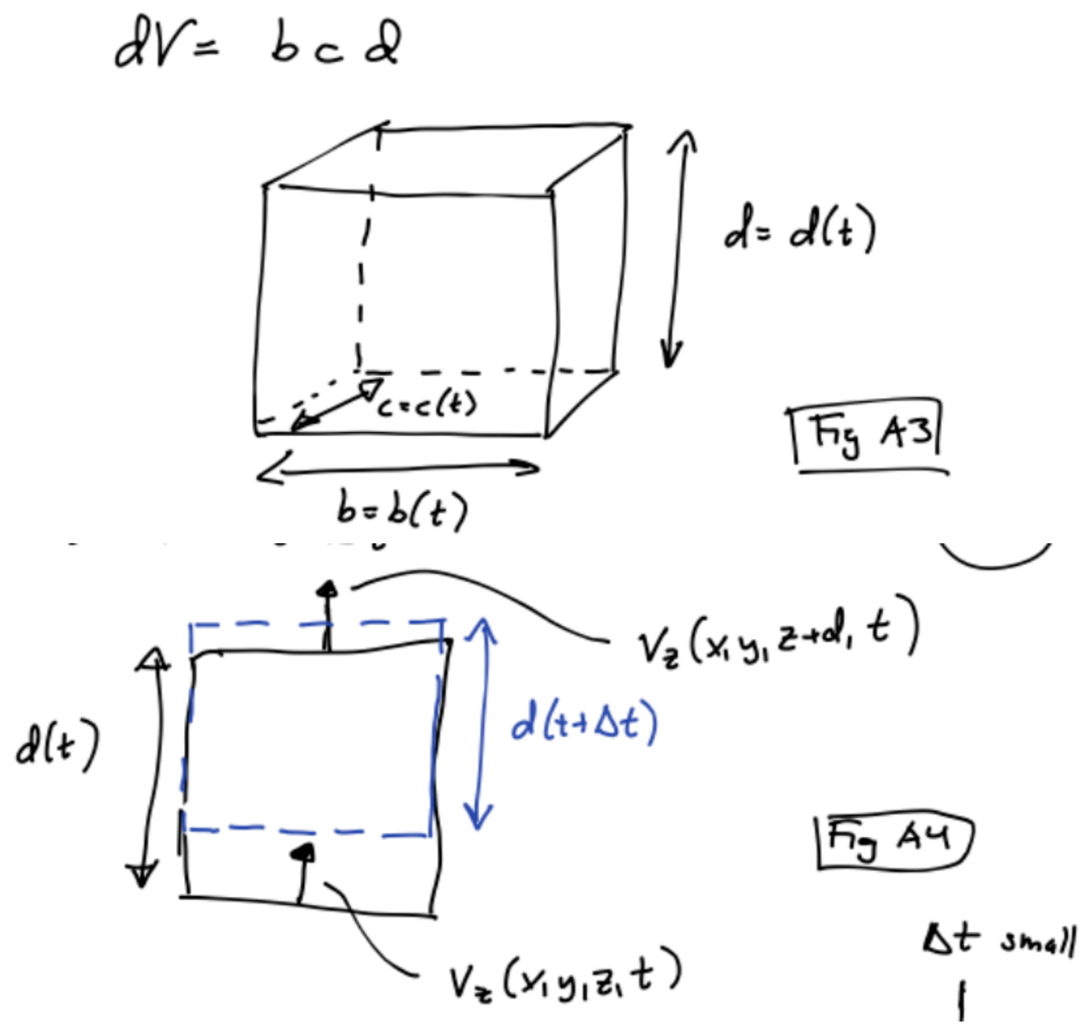


Figure 16: Volume-element derivation of the incompressibility condition.

B.3 The Navier–Stokes equations

We now derive the Navier–Stokes equations.

The derivation proceeds by applying Newton's second law to a small fluid element.

B.3.1 Acceleration of a fluid element

We first derive the acceleration of a fluid element in one dimension. Assume that the velocity field is $v(x, t)$. During a short time interval Δt the fluid element moves

$$\Delta x = v(x, t)\Delta t.$$

The velocity at time $t + \Delta t$ becomes

$$v(x + \Delta x, t + \Delta t).$$

Using a Taylor expansion:

$$v(x + \Delta x, t + \Delta t) = v(x, t) + \frac{\partial v}{\partial t}\Delta t + \frac{\partial v}{\partial x}\Delta x + \mathcal{O}(\Delta t^2).$$

Using $\Delta x = v\Delta t$:

$$v(x + \Delta x, t + \Delta t) = v(x, t) + \left(\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}\right)\Delta t.$$

Hence the acceleration becomes

$$a(x, t) = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}. \tag{129}$$

The second term is called the convective term.

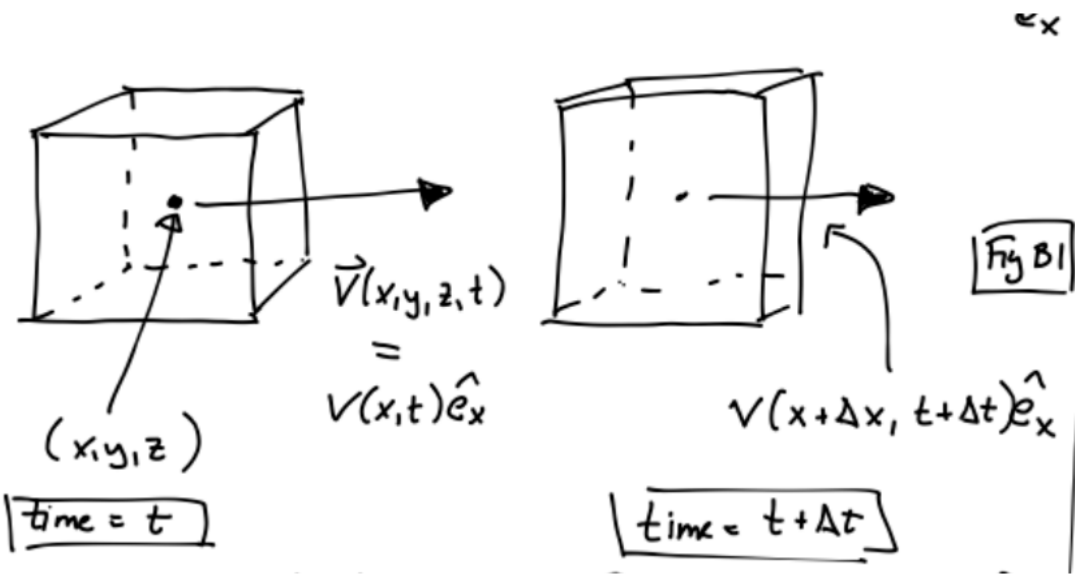


Figure 17: Illustration of a moving fluid element used in the derivation of the convective derivative.

In three dimensions:

$$\mathbf{a}(\mathbf{r}, t) = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v}. \tag{130}$$

B.3.2 Pressure forces

We now derive the pressure force acting on a small volume element.

Again, consider a small box-shaped fluid element with volume

$$dV = bcd.$$

The pressure force in the z-direction is

$$F_z^{(p)} = p(z)bc - p(z + d)bc.$$

Using a Taylor expansion:

$$p(z + d) = p(z) + \frac{\partial p}{\partial z}d + \mathcal{O}(d^2),$$

which gives

$$F_z^{(p)} = -\frac{\partial p}{\partial z} dV.$$

Similarly:

$$F_x^{(p)} = -\frac{\partial p}{\partial x} dV,$$

$$F_y^{(p)} = -\frac{\partial p}{\partial y} dV.$$

Thus:

$$\mathbf{F}^{(p)} = -(\nabla p) dV. \tag{131}$$

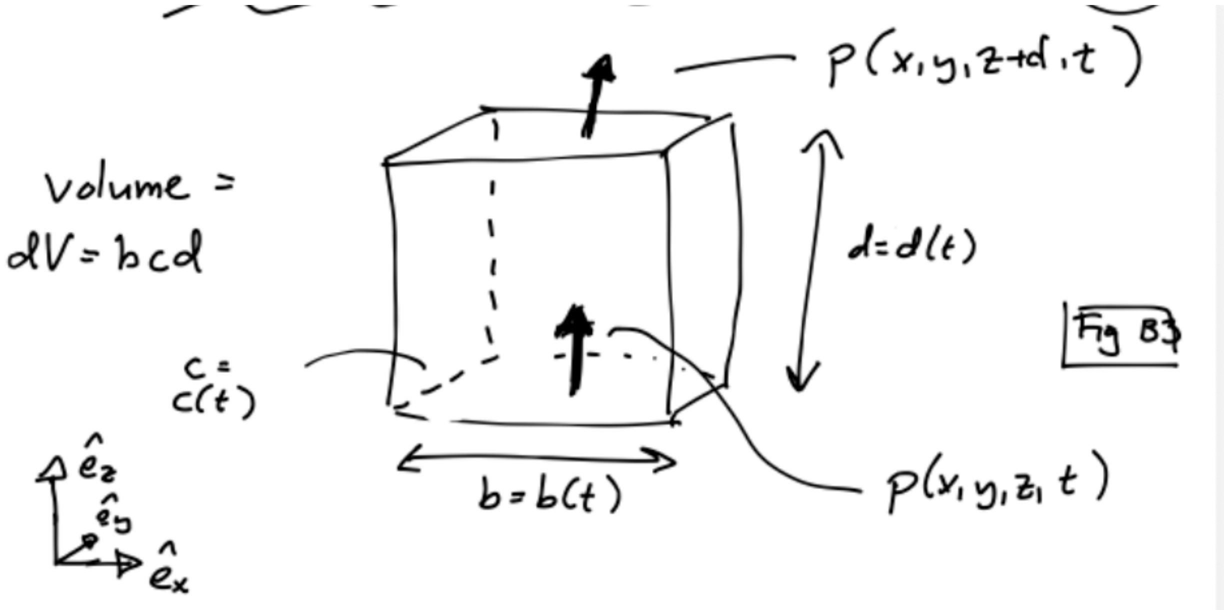


Figure 18: Pressure forces acting on a small fluid element.

B.3.3 Viscous forces

We next derive the viscous force.

Consider first the force contribution in the z-direction due to gradients in the x-direction.

Using Newton's viscosity relation:

$$\frac{F_z}{A} = \eta \frac{\partial v_z}{\partial x}.$$

The force acting on the side located at position x is

$$F_{z,x} = \eta \frac{\partial v_z}{\partial x} cd.$$

The force acting on the opposite side at $x + b$ is

$$F_{z,x+b} = -\eta \left(\frac{\partial v_z}{\partial x} + \frac{\partial^2 v_z}{\partial x^2} b \right) cd.$$

Adding the contributions gives

$$F_z^{(v,x)} = \eta \frac{\partial^2 v_z}{\partial x^2} dV.$$

Repeating the calculation for gradients in the y- and z-directions gives

$$F_z^{(v)} = \eta \left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right) dV.$$

Introducing the Laplacian:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2},$$

we obtain

$$\mathbf{F}^{(v)} = \eta \nabla^2 \mathbf{v} \, dV. \tag{132}$$

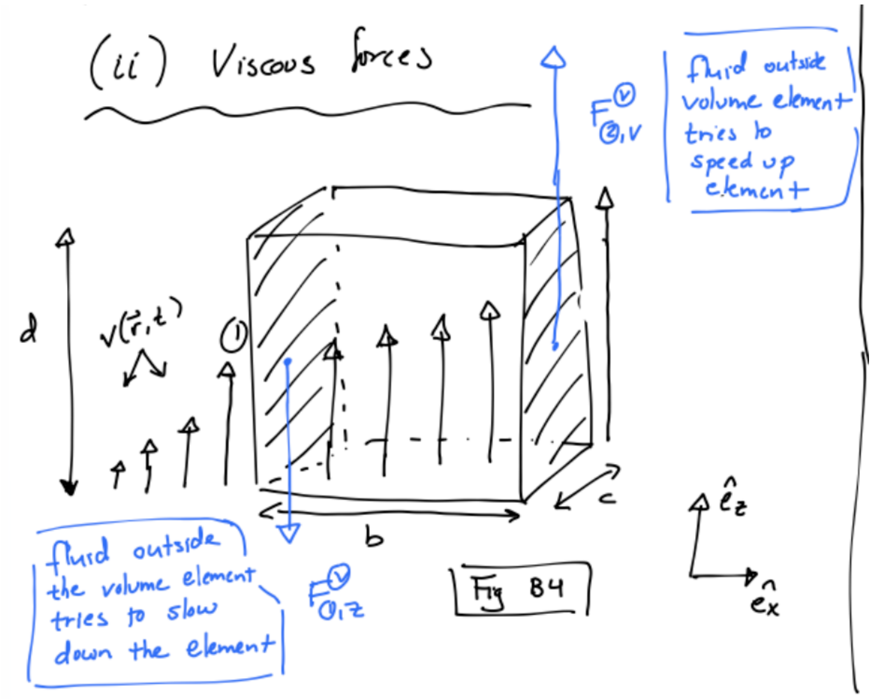


Figure 19: Derivation of viscous forces acting on a fluid element.

B.3.4 The Navier–Stokes equation

We now apply Newton's second law:

$$\mathbf{F} = m\mathbf{a}.$$

The mass of the fluid element is

$$m = \rho dV,$$

where ρ is the mass density.

Using Eqs. (130), (131), and (132), we obtain

$$\rho dV \left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right] = -(\nabla p) dV + \eta \nabla^2 \mathbf{v} \, dV.$$

Dividing by dV yields the Navier–Stokes equation:

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t}+(\mathbf{v}\cdot\nabla)\mathbf{v}\right]=-\nabla p+\eta\nabla^2\mathbf{v}. \tag{133}$$

Together with the incompressibility condition

$$\nabla\cdot\mathbf{v}=0,$$

this equation determines the velocity field.

Finally, the boundary conditions between a liquid and a solid are the non-slip and no penetration ones:

$$\mathbf{v}=\mathbf{v}_s \tag{134}$$

where $\mathbf{v}_s$ is the velocity of the surface.

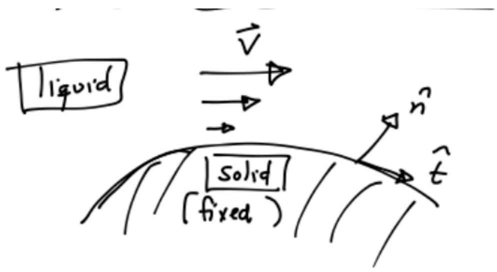


Figure 20: Boundary conditions

C Poiseuille flow – the details

We consider stationary flow through a cylindrical tube of radius R and length L. The pressure difference between the two ends of the tube is Δp. The geometry of the problem is sketched in Fig. 6 in the main text.

We make the following assumptions:

- the flow is stationary,
- the fluid is incompressible,
- the geometry is cylindrically symmetric,
- the flow is only in the z-direction.

Because of these assumptions, the velocity field has the form

$$\mathbf{v}=v_z(r)\hat{\mathbf{z}}.$$

That is, the fluid flows along the tube, but the speed may depend on the radial distance r from the center.

For stationary flow at low Reynolds number, the Navier–Stokes equation reduces to a balance between pressure forces and viscous forces. The z-component becomes

$$0 = -\frac{\partial p}{\partial z} + \eta \nabla^2 v_z.$$

In cylindrical coordinates, for a function depending only on r,

$$\nabla^2 v_z = \frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right).$$

Hence,

$$\frac{\partial p}{\partial z} = \eta \frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right). \tag{135}$$

We assume a constant pressure gradient along the tube,

$$\frac{\partial p}{\partial z} = -\frac{\Delta p}{L}.$$

Equation (135) therefore becomes

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = -\frac{\Delta p}{\eta L}.$$

Multiplying by r gives

$$\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = -\frac{\Delta p}{\eta L} r.$$

We now integrate once:

$$r\frac{dv_z}{dr} = -\frac{\Delta p}{2\eta L} r^2 + C_1.$$

Thus,

$$\frac{dv_z}{dr} = -\frac{\Delta p}{2\eta L} r + \frac{C_1}{r}.$$

The term C_1/r would diverge at the center of the tube, $r = 0$. Since the velocity profile should be finite there, we require

$$C_1 = 0.$$

Hence,

$$\frac{dv_z}{dr} = -\frac{\Delta p}{2\eta L} r.$$

Integrating again gives

$$v_z(r) = -\frac{\Delta p}{4\eta L} r^2 + C_2.$$

We determine C_2 from the no-slip boundary condition at the wall. Since the wall is stationary,

$$v_z(R) = 0.$$

This gives

$$C_2 = \frac{\Delta p}{4\eta L} R^2.$$

Thus,

$$v_z(r) = \frac{\Delta p}{4\eta L}\left(R^2 - r^2\right). \tag{136}$$

The velocity profile is therefore parabolic.

We now calculate the volume flow per unit time,

$$Q = \int v_z(r) dA.$$

Using cylindrical coordinates,

$$dA = 2\pi r\, dr,$$

which gives

$$Q = \int_0^R v_z(r) 2\pi r\, dr.$$

Substituting Eq. (136),

$$Q = \frac{\Delta p}{4\eta L}\int_0^R (R^2 - r^2) 2\pi r\, dr.$$

Evaluating the integral,

$$Q = \frac{\Delta p}{4\eta L} 2\pi \int_0^R (R^2 r - r^3)\, dr.$$

Thus,

$$Q = \frac{\Delta p}{4\eta L} 2\pi \left[\frac{R^2 r^2}{2} - \frac{r^4}{4}\right]_0^R.$$

This gives

$$Q = \frac{\Delta p}{4\eta L} 2\pi \left[\frac{R^4}{2} - \frac{R^4}{4}\right] = \frac{\pi R^4}{8\eta L}\Delta p.$$

Thus,

$$Q = \frac{\pi R^4}{8\eta L}\Delta p. \tag{137}$$

This result is known as the Hagen–Poiseuille law. The most important feature is the scaling

$$Q \propto R^4.$$

This fourth-power dependence is dramatic. If the radius of a blood vessel is reduced by a factor of two, the flow rate is reduced by a factor of $2^4 = 16$, for the same pressure difference. Conversely, a modest increase in vessel radius can strongly increase blood flow.

D Solving ordinary differential equations numerically

Many of the models in these notes lead to ordinary differential equations. For example, rate equations have the general form

$$\frac{dC_i}{dt} = F_i(C_1, C_2, \ldots; k_1, k_2, \ldots), \tag{138}$$

where the C_i are concentrations and the k_i are rate constants.

In vector notation we write such systems as

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y}), \qquad \mathbf{y}(t_0) = \mathbf{y}_0. \tag{139}$$

Here

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \end{pmatrix}$$

contains all dynamical variables.

For example, in the full Michaelis–Menten model we may choose

$$\mathbf{y}(t) = \begin{pmatrix} [E](t) \\ [S](t) \\ [ES](t) \\ [P](t) \end{pmatrix}.$$

The goal of a numerical ODE solver is simple: given $\mathbf{y}(t_0)$, use Eq. (139) to compute approximate values of $\mathbf{y}(t)$ at later times.

The basic idea is to discretize time,

$$t_n = t_0 + nh, \qquad n = 0, 1, 2, \ldots, \tag{140}$$

where h is the time step, and then approximate

$$\mathbf{y}(t_n)$$

by a numerical value

$$\mathbf{y}_n.$$

D.1 Euler's method

The simplest method is Euler's method.

Starting from

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y}),$$

we approximate the derivative by a forward difference,

$$\left.\frac{d\mathbf{y}}{dt}\right|_{t=t_n} \approx \frac{\mathbf{y}_{n+1} - \mathbf{y}_n}{h}.$$

Inserting this into Eq. (139) gives

$$\frac{\mathbf{y}_{n+1} - \mathbf{y}_n}{h} \approx \mathbf{f}(t_n, \mathbf{y}_n).$$

Solving for $\mathbf{y}_{n+1}$, we obtain the Euler update rule

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h\mathbf{f}(t_n, \mathbf{y}_n). \tag{141}$$

This is the method used in the simple numerical examples in the main text. The interpretation is very physical. At time t_n, we evaluate the velocity, or rate of change,

$$\mathbf{f}(t_n, \mathbf{y}_n),$$

and assume that this rate remains approximately constant during the next small time interval of length h.

Thus,

$$\text{new value} = \text{old value} + \text{time step} \times \text{rate of change}.$$

A scalar example is

$$\frac{dy}{dt} = -ky.$$

Euler's method gives

$$y_{n+1} = y_n - hky_n = (1 - hk)y_n.$$

This should remind us of the microscopic decay argument used earlier in the notes.

A minimal Python implementation is:

```
for n in range(N):
    y = y + h*f(t, y)
    t = t + h
```

For vector-valued problems, the same idea applies. If `y` is a NumPy array, then `f(t,y)` should return a NumPy array of the same size:

```
for n in range(N):
    y = y + h*f(t, y)
    t = t + h
```

The code looks identical; only now `y` represents several variables.

D.2 Example: Euler method for Michaelis–Menten kinetics

Consider the enzyme reaction

$$E + S \underset{k_-}{\overset{k_+}{\rightleftarrows}} ES \xrightarrow{r} E + P.$$

The full rate equations are

$$
\begin{aligned}
\frac{d[E]}{dt} &= -k_+[E][S] + k_-[ES] + r[ES], && (142)\\
\frac{d[S]}{dt} &= -k_+[E][S] + k_-[ES], && (143)\\
\frac{d[ES]}{dt} &= k_+[E][S] - k_-[ES] - r[ES], && (144)\\
\frac{d[P]}{dt} &= r[ES]. && (145)
\end{aligned}
$$

Let

$$\mathbf{y} = \begin{pmatrix} [E] \\ [S] \\ [ES] \\ [P] \end{pmatrix}.$$

Then the right-hand side $\mathbf{f}(\mathbf{y})$ is

$$\mathbf{f}(\mathbf{y}) = \begin{pmatrix} -k_+[E][S] + k_-[ES] + r[ES] \\ -k_+[E][S] + k_-[ES] \\ k_+[E][S] - k_-[ES] - r[ES] \\ r[ES] \end{pmatrix}.$$

A compact Python implementation is:

```
import numpy as np
import matplotlib.pyplot as plt

def f(t, y, k_plus, k_minus, r):
    E, S, ES, P = y

    dE  = -k_plus*E*S + k_minus*ES + r*ES
    dS  = -k_plus*E*S + k_minus*ES
    dES =  k_plus*E*S - k_minus*ES - r*ES
    dP  =  r*ES

    return np.array([dE, dS, dES, dP])

# parameters
k_plus  = 1.0
k_minus = 1.0
r       = 0.5

# initial condition: [E], [S], [ES], [P]
y = np.array([1.0, 10.0, 0.0, 0.0])
```

```
# time grid
h = 1e-3
T = 20.0
N = int(T/h)

t_values = np.zeros(N+1)
y_values = np.zeros((N+1, 4))

y_values[0, :] = y

t = 0.0
for n in range(N):
    y = y + h*f(t, y, k_plus, k_minus, r)
    t = t + h

    t_values[n+1] = t
    y_values[n+1, :] = y

plt.plot(t_values, y_values[:,0], label='E')
plt.plot(t_values, y_values[:,1], label='S')
plt.plot(t_values, y_values[:,2], label='ES')
plt.plot(t_values, y_values[:,3], label='P')
plt.xlabel('time')
plt.ylabel('concentration')
plt.legend()
plt.show()
```

This code is deliberately simple. It is not meant to be the best possible ODE solver. It is meant to show directly how the differential equations re turned into a numerical time evolution.

The same structure can be used in Matlab using arrays and a loop. In C++, one would usually store the components of $\mathbf{y}$ in a vector or array and update them explicitly.

D.3 Local versus global error

Euler's method is approximate. To understand the error, consider the exact solution expanded in a Taylor series:

$$\mathbf{y}(t_n + h) = \mathbf{y}(t_n) + h\frac{d\mathbf{y}}{dt}\bigg|_{t=t_n} + \frac{h^2}{2}\frac{d^2\mathbf{y}}{dt^2}\bigg|_{t=t_n} + \mathcal{O}(h^3).$$

Using

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y}),$$

this becomes

$$\mathbf{y}(t_n + h) = \mathbf{y}(t_n) + h\mathbf{f}(t_n, \mathbf{y}(t_n)) + \mathcal{O}(h^2).$$

Euler's method keeps the first two terms and neglects terms of order h^2 and higher. Thus the error made in a single step, called the *local truncation error*, is of order

$$h^2.$$

However, we usually integrate for many steps. If we integrate to a fixed final time

$$T = Nh,$$

then the number of steps is

$$N = \frac{T}{h}.$$

Very roughly, the local errors accumulate over N steps. Therefore the global error scales as

$$Nh^2 = \frac{T}{h}h^2 = Th.$$

Thus Euler's method has global error

$$\text{global error} \sim h. \tag{146}$$

Euler's method is therefore called a first-order method.

More generally, if the local error scales as

$$h^m,$$

then the global error typically scales as

$$h^{m-1}. \tag{147}$$

This distinction between local and global error is important. A method may look quite accurate over one step but still accumulate significant error over many steps.

D.4 Step-size checks

A practical rule is:

never trust a numerical solution until you have changed the step size.

For example, compute the solution using

$$h, \qquad \frac{h}{2}, \qquad \frac{h}{4}.$$

If the result changes significantly, then the time step is too large.

For Euler's method, halving the step size should approximately halve the global error, provided the time step is already small enough that the asymptotic error scaling applies.

In biological rate equations, another useful check is conservation. For example, in the Michaelis–Menten system,

$$[E] + [ES]$$

should remain constant, because enzyme is neither created nor destroyed. Similarly,

$$[S] + [ES] + [P]$$

should remain constant for the simple reaction scheme. If these quantities drift significantly in a simulation, something is wrong: either the code has an error, or the time step is too large.

D.5 Stability

Accuracy is not the only issue. A numerical method must also be stable.

Consider again the simple decay equation

$$\frac{dy}{dt} = -ky.$$

Euler's method gives

$$y_{n+1} = (1 - hk)y_n.$$

The exact solution decays smoothly to zero. For the numerical solution to decay rather than grow, we need

$$|1 - hk| < 1.$$

This gives

$$0 < hk < 2. \tag{148}$$

If h is too large, the numerical solution can oscillate or even blow up, even though the exact solution is perfectly well behaved.

This is especially important for rate equations with fast and slow time scales. The fastest rate in the problem often sets the maximum safe time step.

A useful rough rule is

$$h \ll \frac{1}{k_{\text{fast}}},$$

where k_{fast} is the largest relevant rate constant.

D.6 The Runge–Kutta idea

Euler's method uses the slope at the beginning of the interval:

$$\mathbf{f}(t_n, \mathbf{y}_n).$$

The basic idea of Runge–Kutta methods is to estimate the change in $\mathbf{y}$ using slopes evaluated at several carefully chosen points inside the interval. This usually gives much better accuracy for the same step size.

D.7 Second-order Runge–Kutta method

A common second-order Runge–Kutta method is the midpoint method. It is defined by

$$\mathbf{k}_1 = h\mathbf{f}(t_n, \mathbf{y}_n), \tag{149}$$

$$\mathbf{k}_2 = h\mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{\mathbf{k}_1}{2}\right), \tag{150}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \mathbf{k}_2. \tag{151}$$

The interpretation is simple. We first use Euler's method to estimate the state halfway through the time step. We then evaluate the slope at this midpoint and use that slope to update the solution over the full step.

The local error of this method is of order

$$h^3,$$

so the global error is of order

$$h^2.$$

Thus the midpoint method is a second-order method.

A minimal Python implementation is:

```
for n in range(N):
    k1 = h*f(t, y)
    k2 = h*f(t + h/2, y + k1/2)

    y = y + k2
    t = t + h
```

This is only slightly more complicated than Euler's method, but often much more accurate.

D.8 General second-order Runge–Kutta family

The midpoint method is one member of a broader family. Consider the ansatz

$$\mathbf{k}_1 = h\mathbf{f}(t_n, \mathbf{y}_n), \tag{152}$$

$$\mathbf{k}_2 = h\mathbf{f}(t_n + mh, \mathbf{y}_n + m\mathbf{k}_1), \tag{153}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + a\mathbf{k}_1 + b\mathbf{k}_2. \tag{154}$$

The constants a, b, and m are chosen so that the Taylor expansion of the numerical update agrees with the exact solution as far as possible.

For scalar y, the exact solution satisfies

$$\begin{aligned} y(t_n + h) = {} & y_n + hf(t_n, y_n) \\ & + \frac{h^2}{2}\left[\frac{\partial f}{\partial t}(t_n, y_n) + f(t_n, y_n)\frac{\partial f}{\partial y}(t_n, y_n)\right] + \mathcal{O}(h^3). \end{aligned} \tag{155}$$

Expanding the Runge–Kutta ansatz gives

$$\begin{aligned} y_{n+1} &= y_n + (a+b)hf(t_n, y_n) \\ &\quad + bmh^2 \left[\frac{\partial f}{\partial t}(t_n, y_n) + f(t_n, y_n) \frac{\partial f}{\partial y}(t_n, y_n) \right] + \mathcal{O}(h^3). \end{aligned} \tag{156}$$

Comparing Eqs. (155) and (156), we obtain the conditions

$$a + b = 1, \qquad bm = \frac{1}{2}. \tag{157}$$

These conditions define a one-parameter family of second-order Runge–Kutta methods.

For example, choosing

$$m = \frac{1}{2}, \qquad b = 1, \qquad a = 0,$$

gives the midpoint method above.

Choosing

$$m = 1, \qquad b = \frac{1}{2}, \qquad a = \frac{1}{2},$$

gives Heun's method,

$$\mathbf{k}_1 = h\mathbf{f}(t_n, \mathbf{y}_n), \tag{158}$$

$$\mathbf{k}_2 = h\mathbf{f}(t_n + h, \mathbf{y}_n + \mathbf{k}_1), \tag{159}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{1}{2}(\mathbf{k}_1 + \mathbf{k}_2). \tag{160}$$

Both are second-order methods.

D.9 Fourth-order Runge–Kutta

A very widely used method is the classical fourth-order Runge–Kutta method, often called RK4. It is given by

$$\mathbf{k}_1 = h\mathbf{f}(t_n, \mathbf{y}_n), \tag{161}$$

$$\mathbf{k}_2 = h\mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{\mathbf{k}_1}{2}\right), \tag{162}$$

$$\mathbf{k}_3 = h\mathbf{f}\left(t_n + \frac{h}{2}, \mathbf{y}_n + \frac{\mathbf{k}_2}{2}\right), \tag{163}$$

$$\mathbf{k}_4 = h\mathbf{f}\left(t_n + h, \mathbf{y}_n + \mathbf{k}_3\right), \tag{164}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{1}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \tag{165}$$

The global error of RK4 scales as

$$h^4.$$

This method is often a very good default choice for smooth ODEs.

In Python one could write:

```
for n in range(N):
```

```
k1 = h*f(t, y)
k2 = h*f(t + h/2, y + k1/2)
k3 = h*f(t + h/2, y + k2/2)
k4 = h*f(t + h,   y + k3)

y = y + (k1 + 2*k2 + 2*k3 + k4)/6
t = t + h
```

In actual research code, one often uses built-in adaptive ODE solvers, for example `scipy.integrate.solve_ivp` in Python or `ode45` in Matlab. However, it is still very useful to understand Euler and Runge–Kutta methods, because they reveal what the computer is actually doing.